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Lowe et al.

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(54) **BASE STATIONS INCLUDING INTEGRATED SYSTEMS FOR SERVICING UAVS**

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B60L 53/00 (2019.01)
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B64U 70/50 (2023.01)
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B64U 80/10 (2023.01)
H01M 10/44 (2006.01)
H01M 10/613 (2014.01)
H01M 10/617 (2014.01)
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H01M 10/63 (2014.01)
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H01M 10/6556 (2014.01)
H01M 10/6566 (2014.01)
H01M 10/6572 (2014.01)
B64U 50/19 (2023.01)
B64U 70/80 (2023.01)
B64U 80/70 (2023.01)

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(2013.01); **B64U 70/95** (2023.01); **B64U 80/10** (2023.01); **H01M 10/443** (2013.01); **H01M 10/613** (2015.04); **H01M 10/617** (2015.04); **H01M 10/625** (2015.04); **H01M 10/63** (2015.04); **H01M 10/653** (2015.04); **H01M 10/6551** (2015.04); **H01M 10/6556** (2015.04); **H01M 10/6566** (2015.04); **H01M 10/6572** (2015.04); **B60L 2200/10** (2013.01); **B64U 50/19** (2023.01); **B64U 70/80** (2023.01); **B64U 80/70** (2023.01); **H01M 2220/20** (2013.01)

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See application file for complete search history.

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(57) **ABSTRACT**

An unmanned aerial vehicle (UAV) is disclosed that includes a power source. The power source includes: one or more power cells; one or more thermal transfer members that are thermally connected to the one or more power cells; and a heat exchanger that is thermally connected to the one or more thermal transfer members such that the one or more thermal transfer members and the heat exchanger facilitate a transfer of thermal energy between the power source and ambient air to decrease or increase temperature of the power source.

20 Claims, 18 Drawing Sheets

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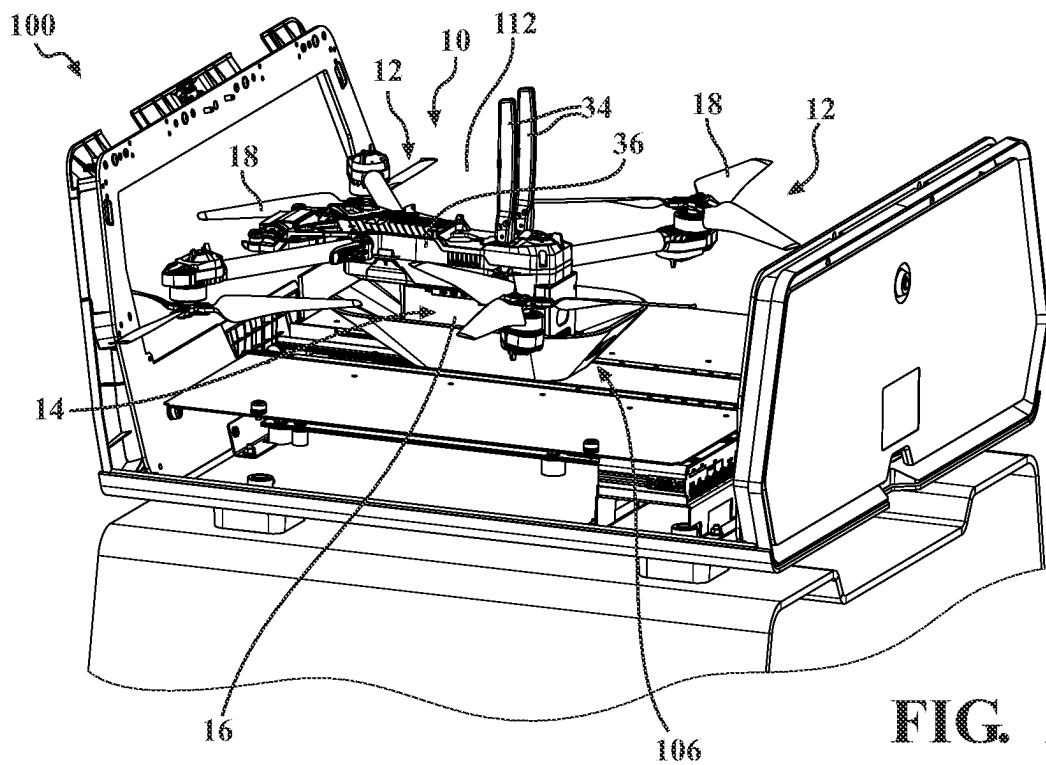
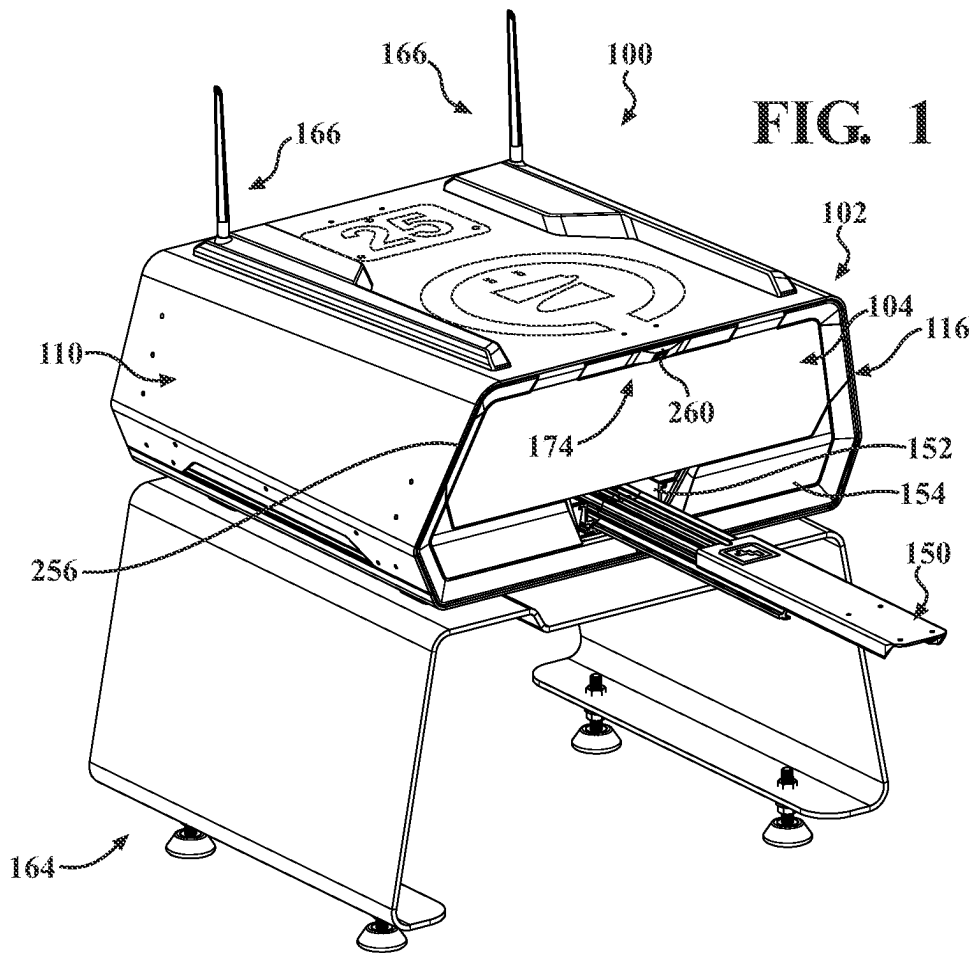
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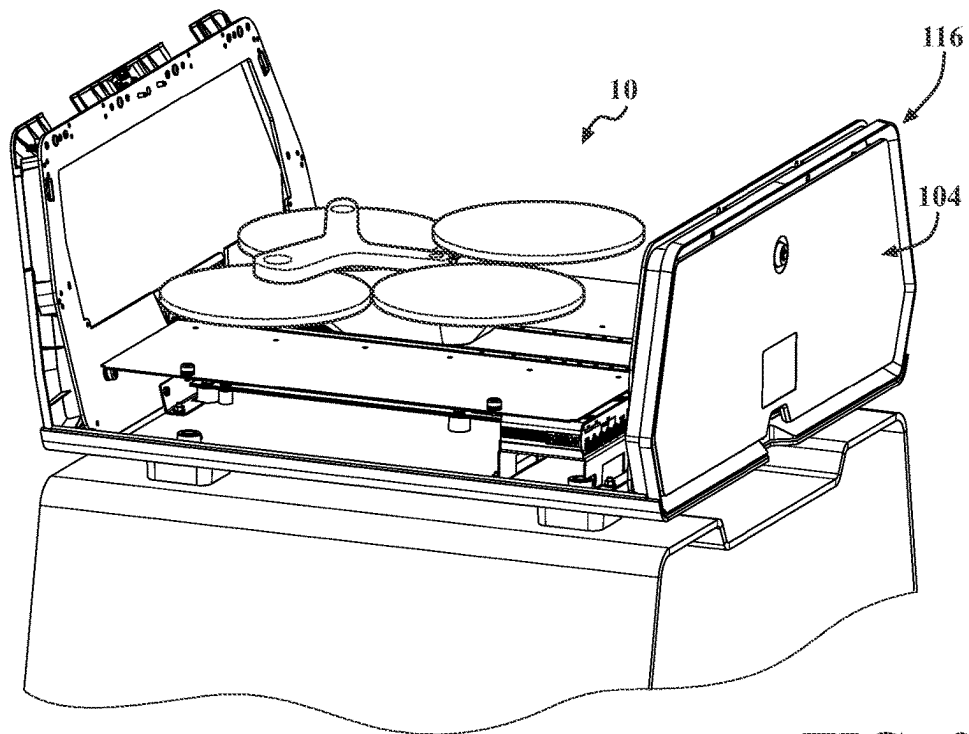


FIG. 3

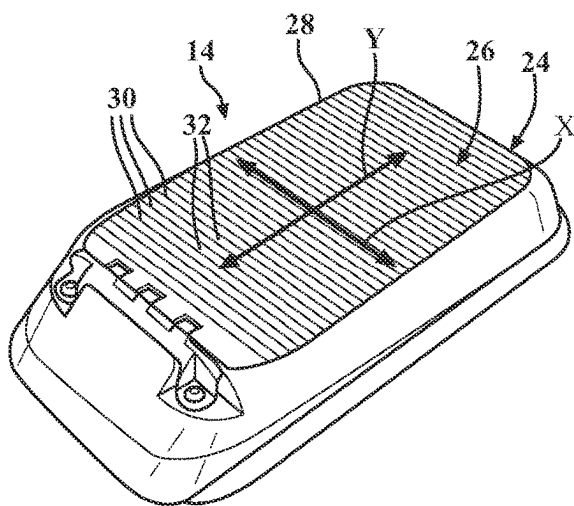


FIG. 4

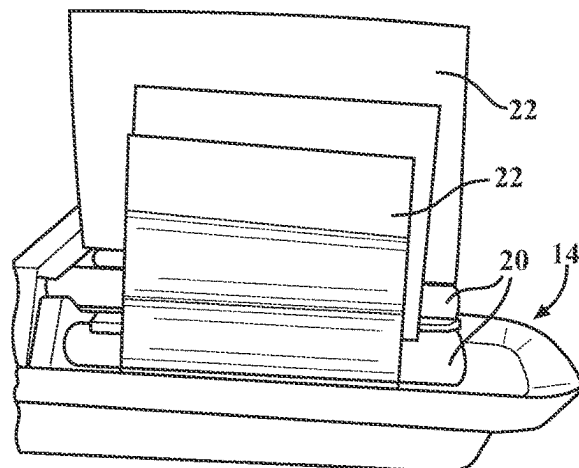


FIG. 5

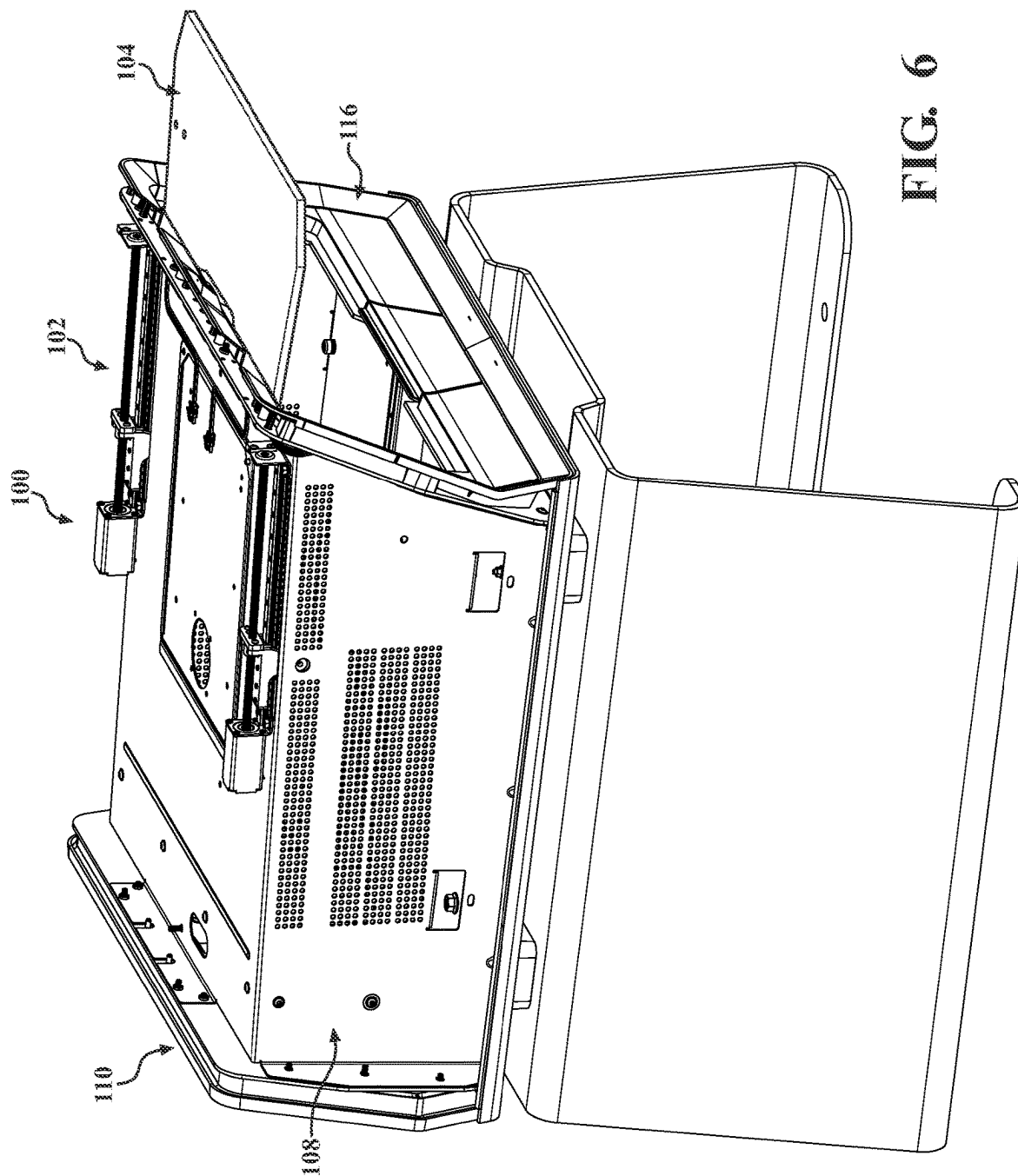


FIG. 6

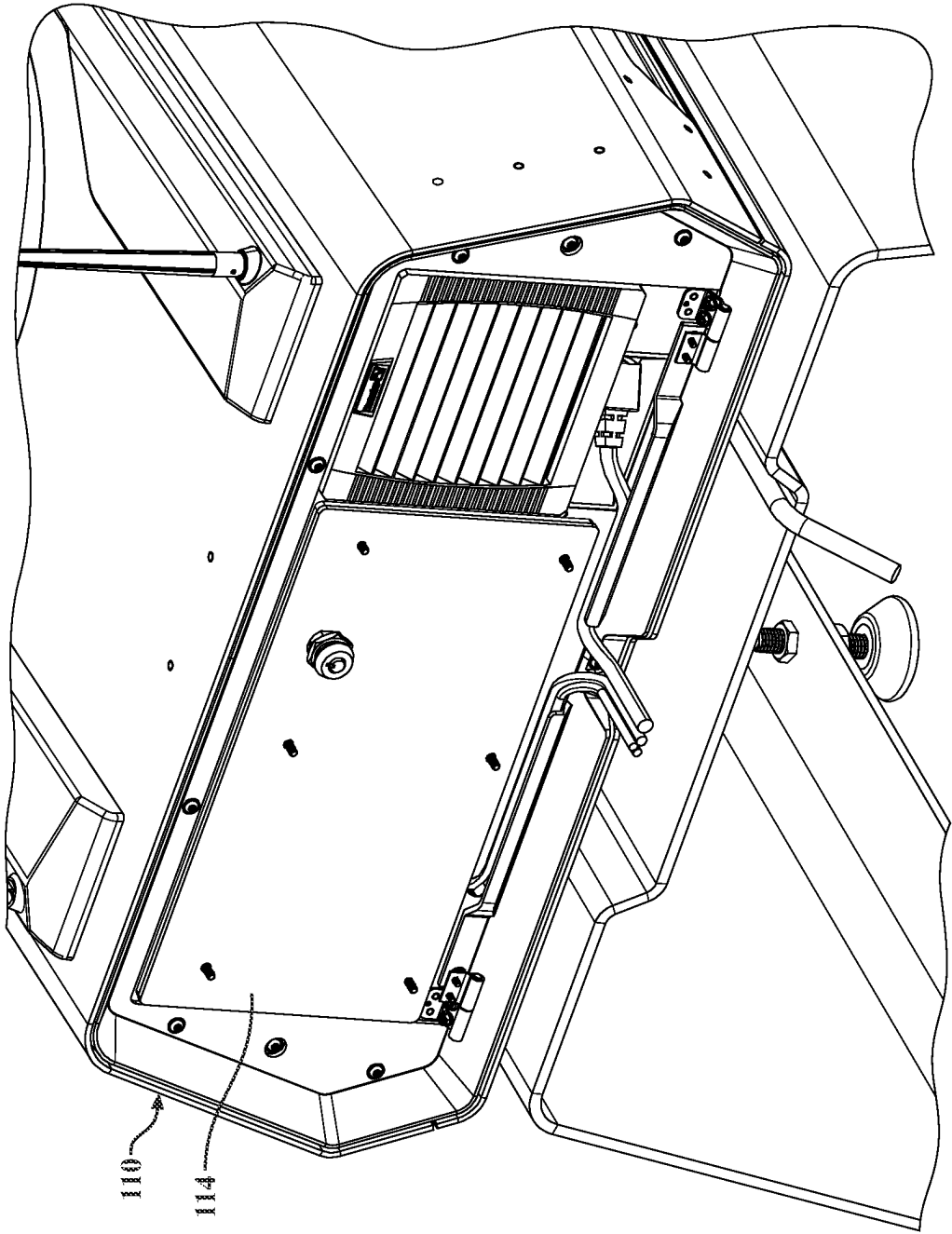


FIG. 7

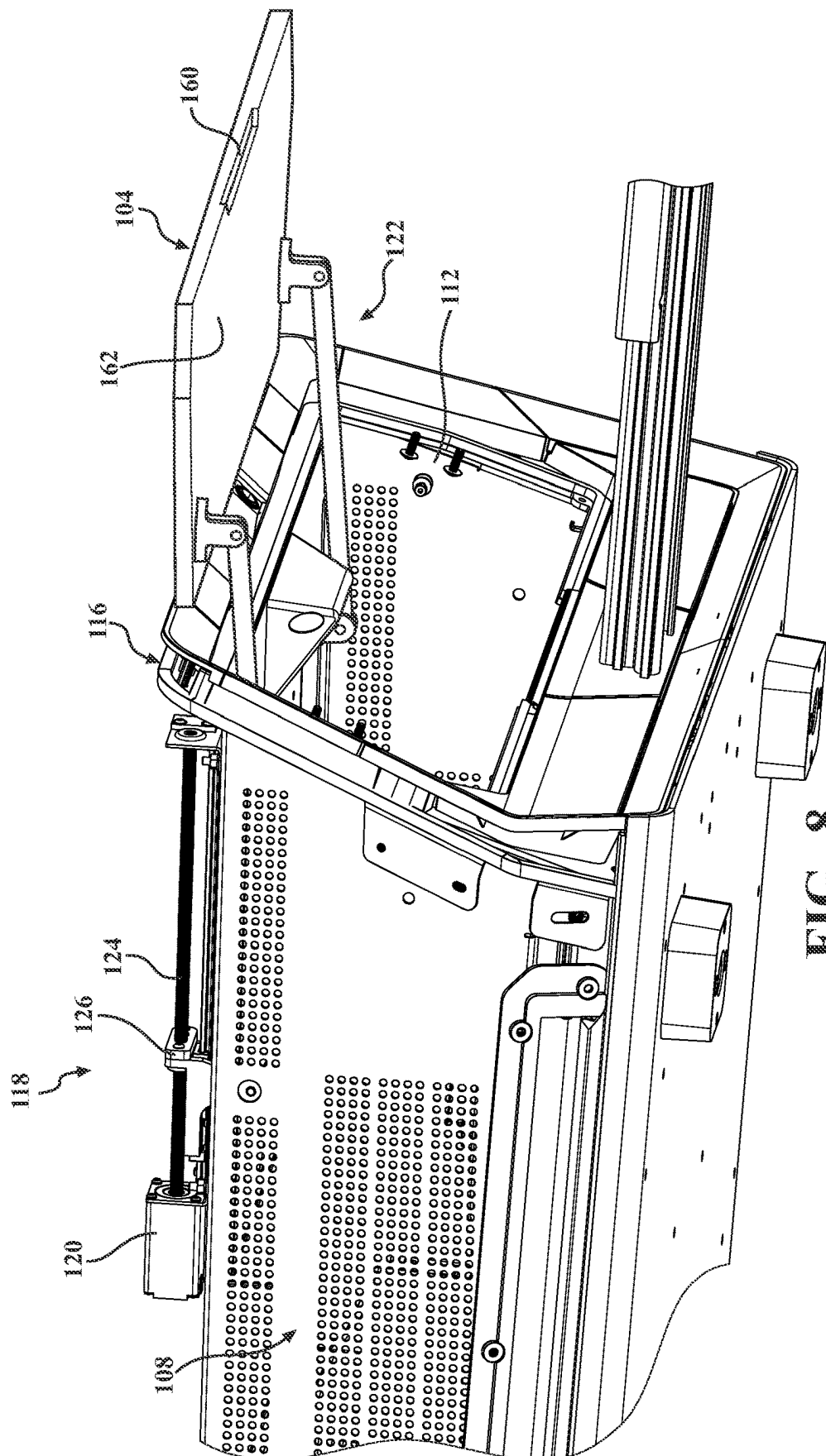


FIG 8

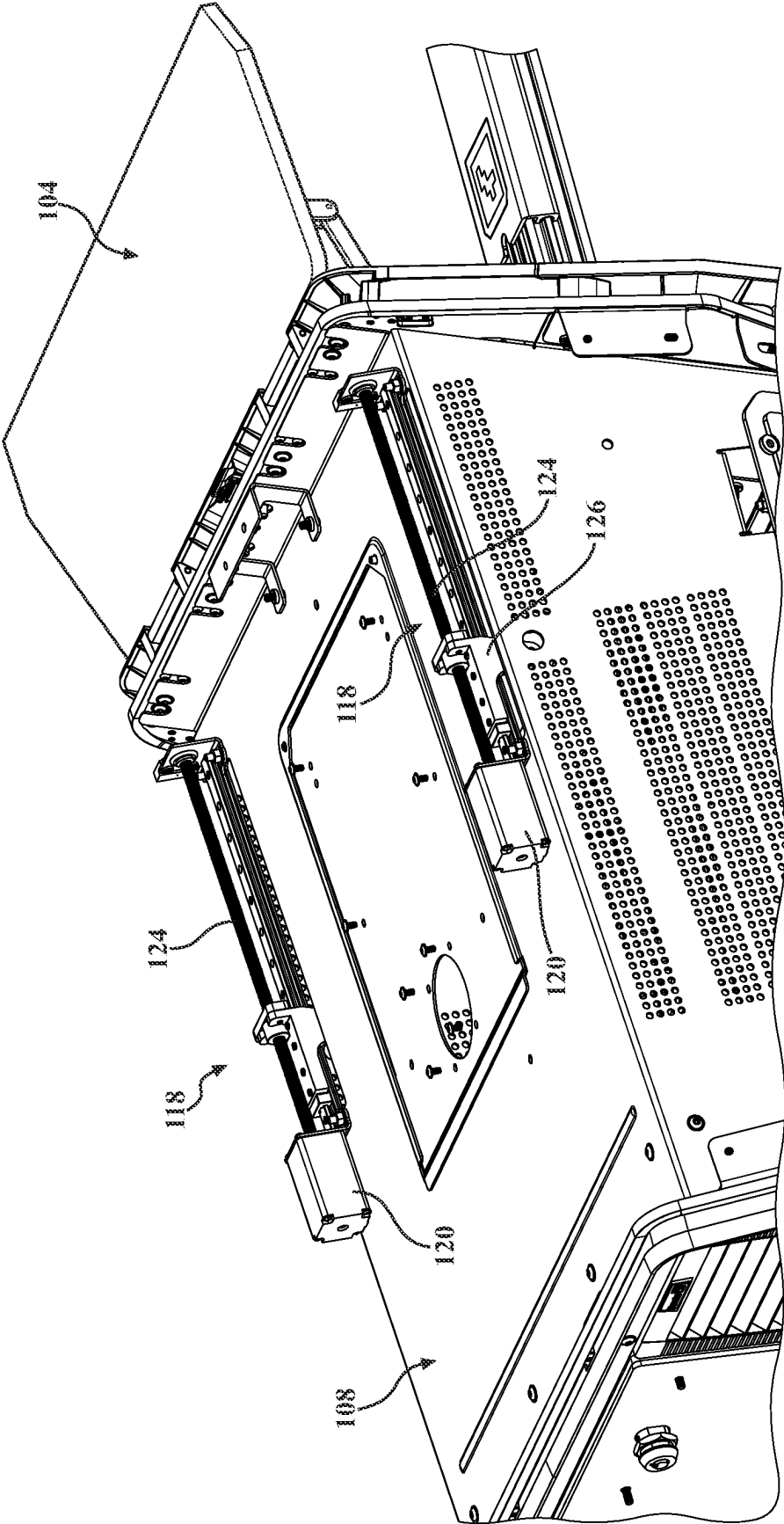


FIG. 9

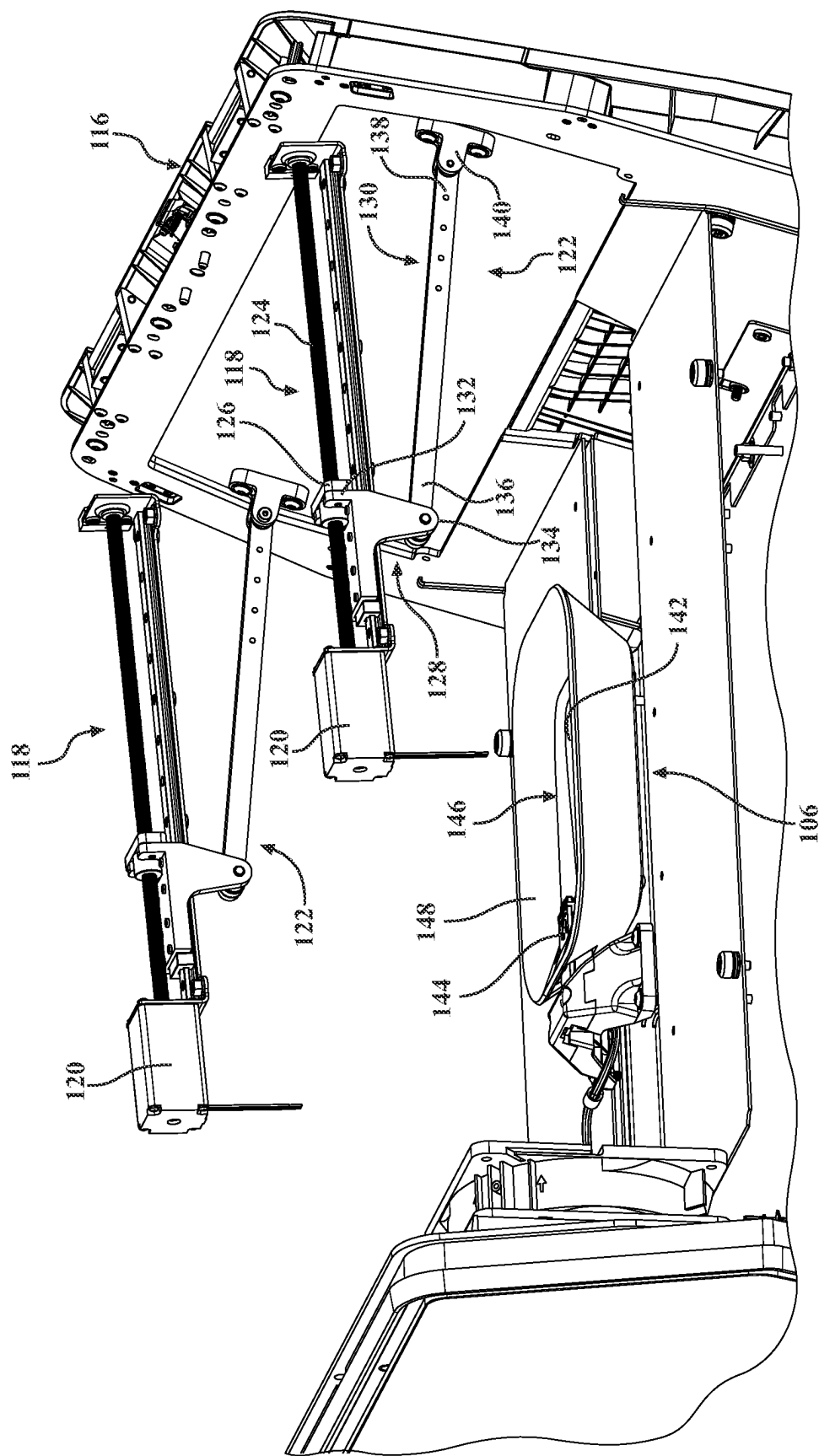


FIG. 10

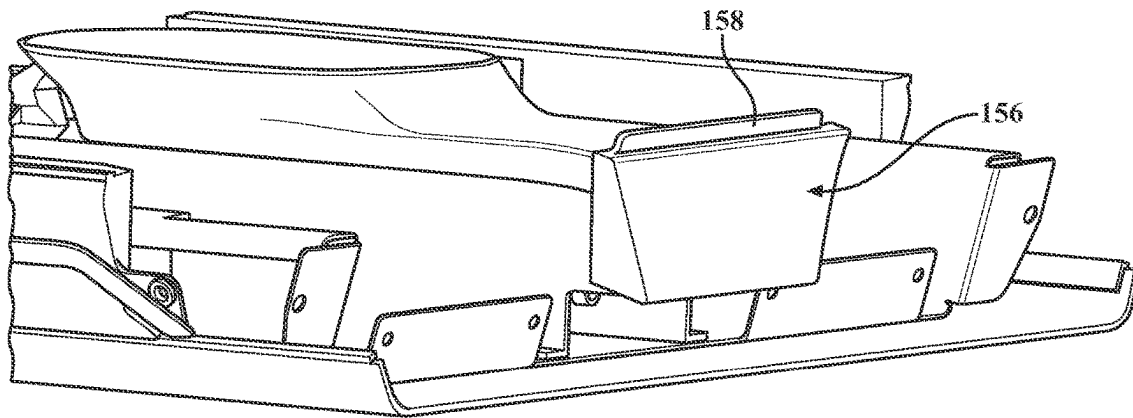


FIG. 11

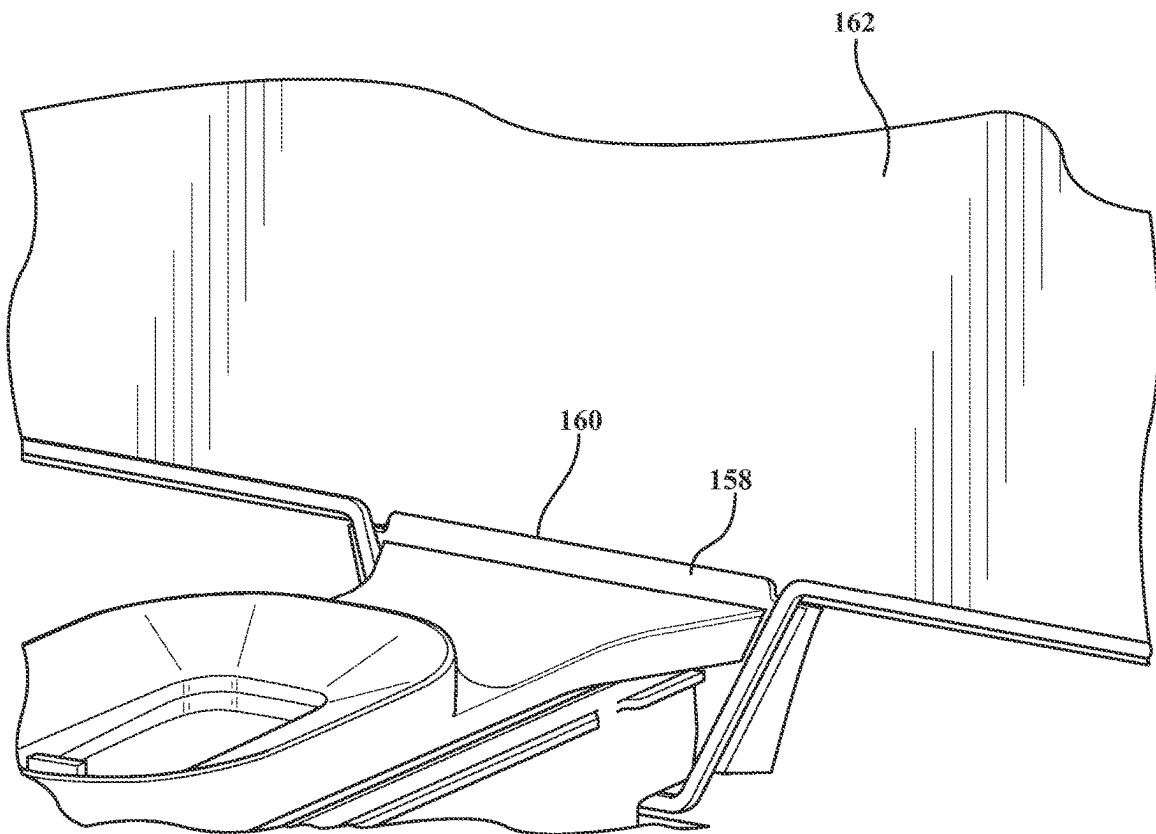


FIG. 12

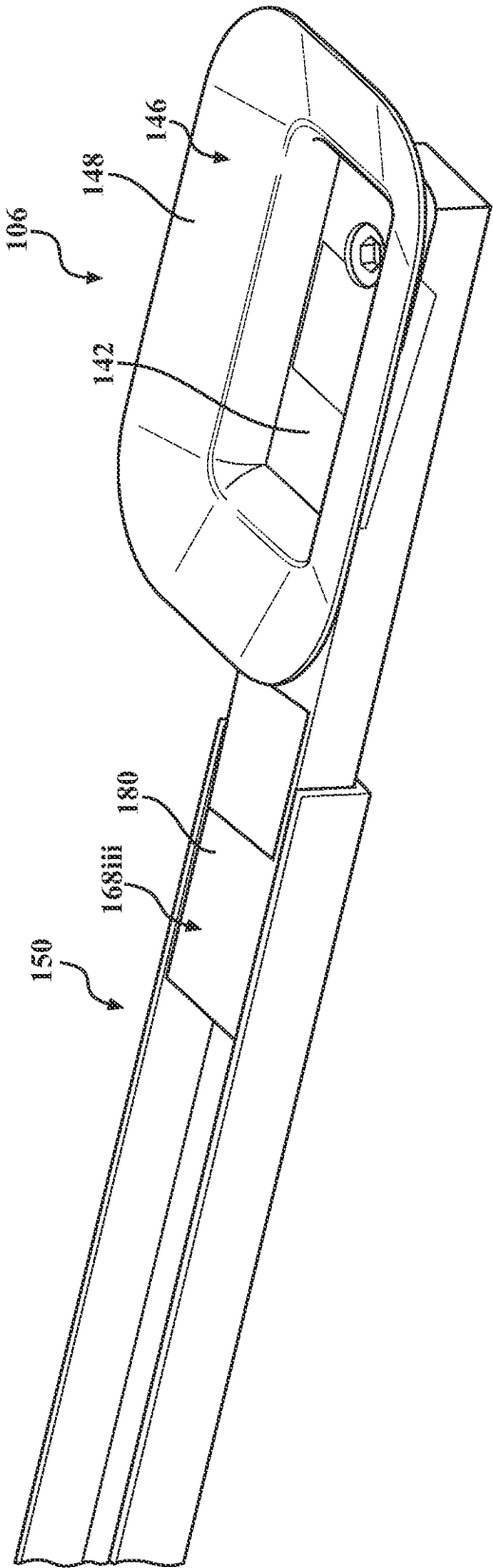


FIG. 13

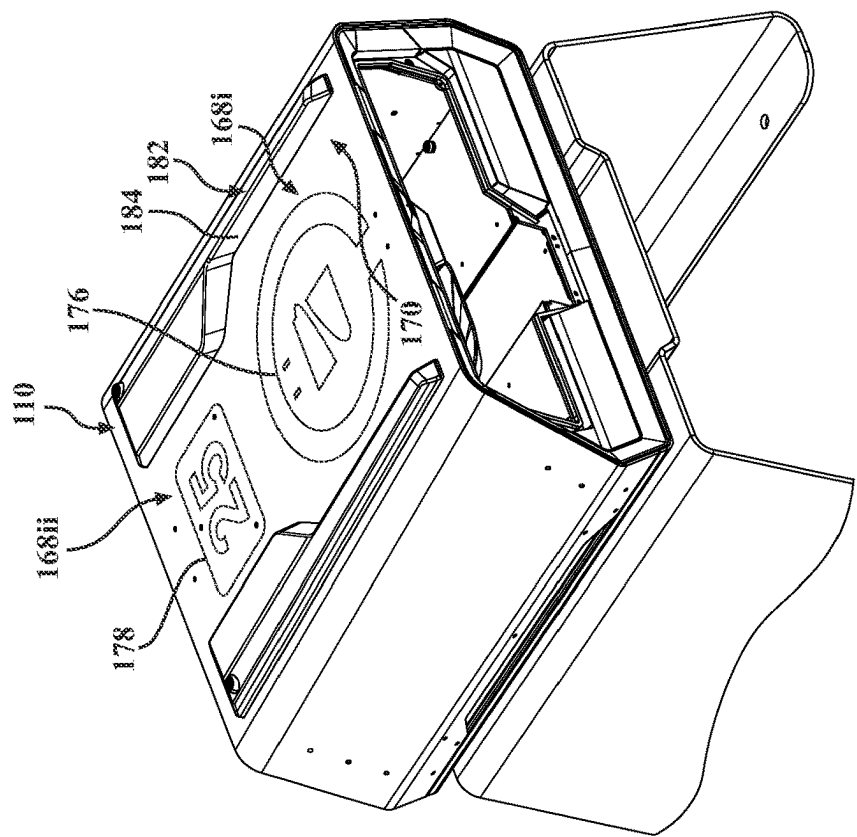


FIG. 15

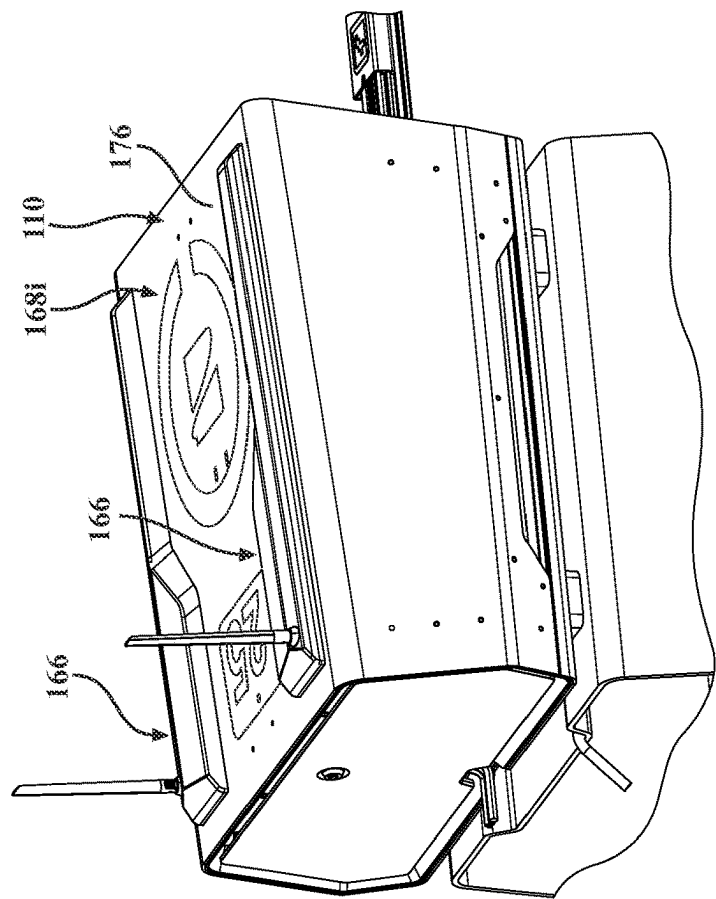


FIG. 14

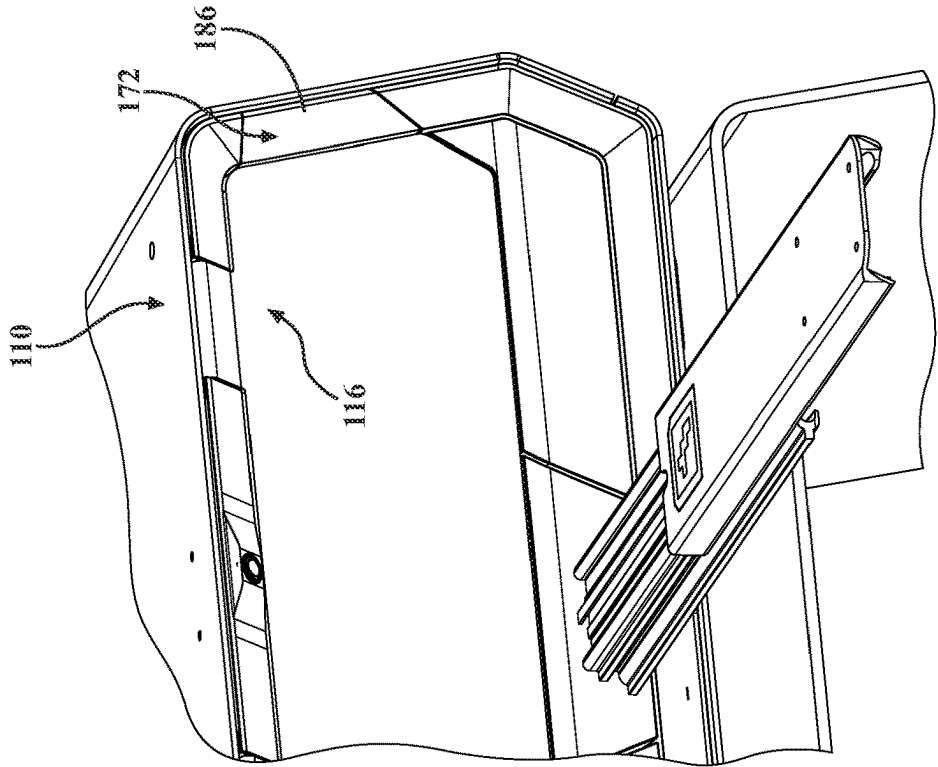
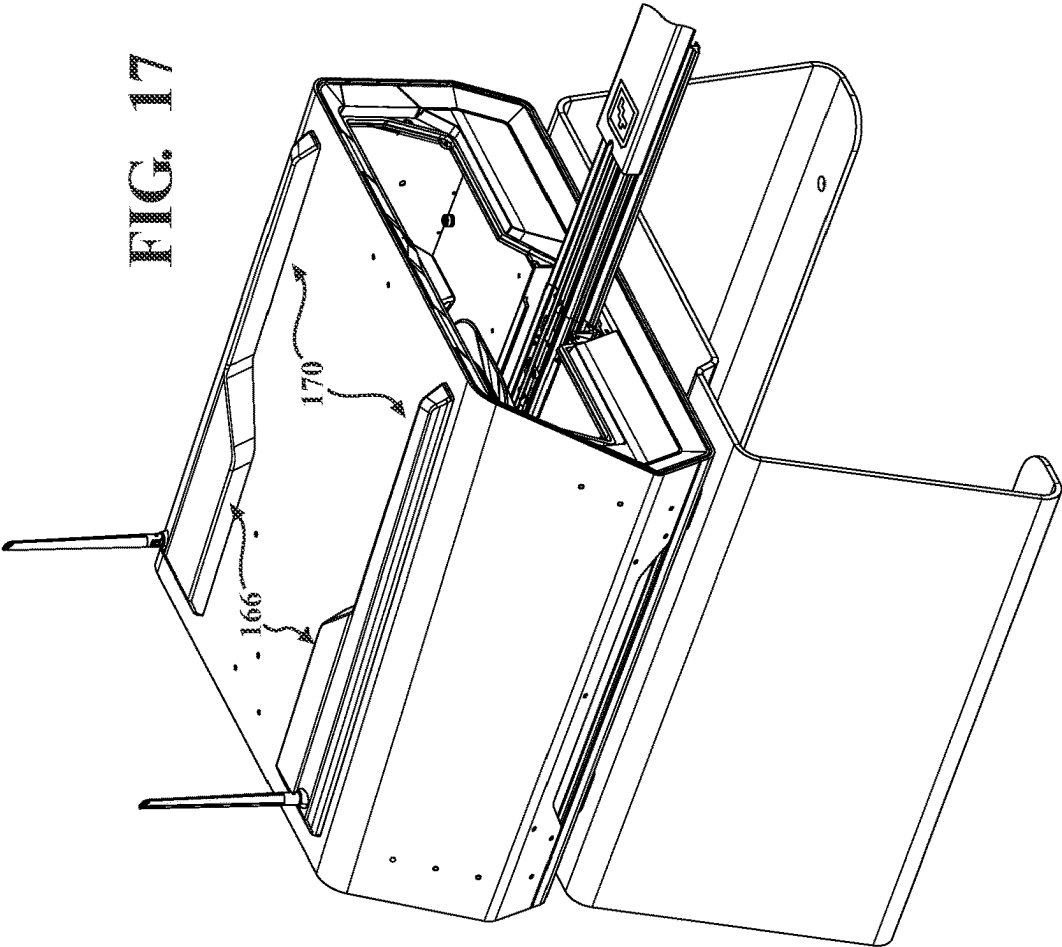


FIG. 16

FIG. 17

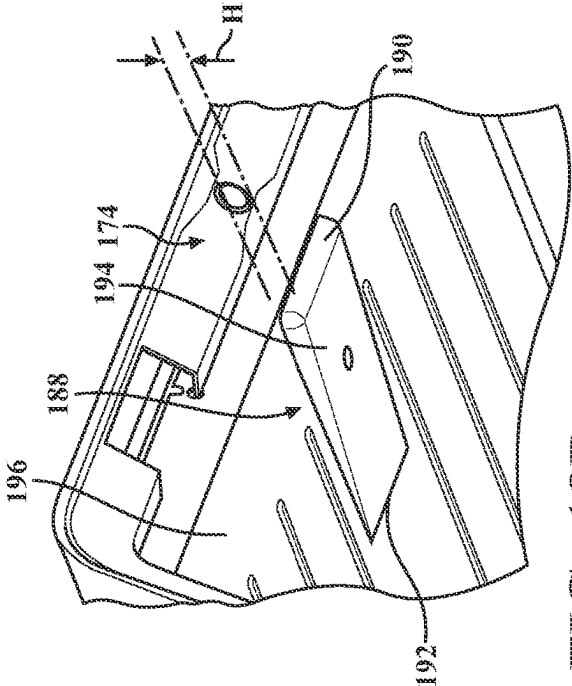
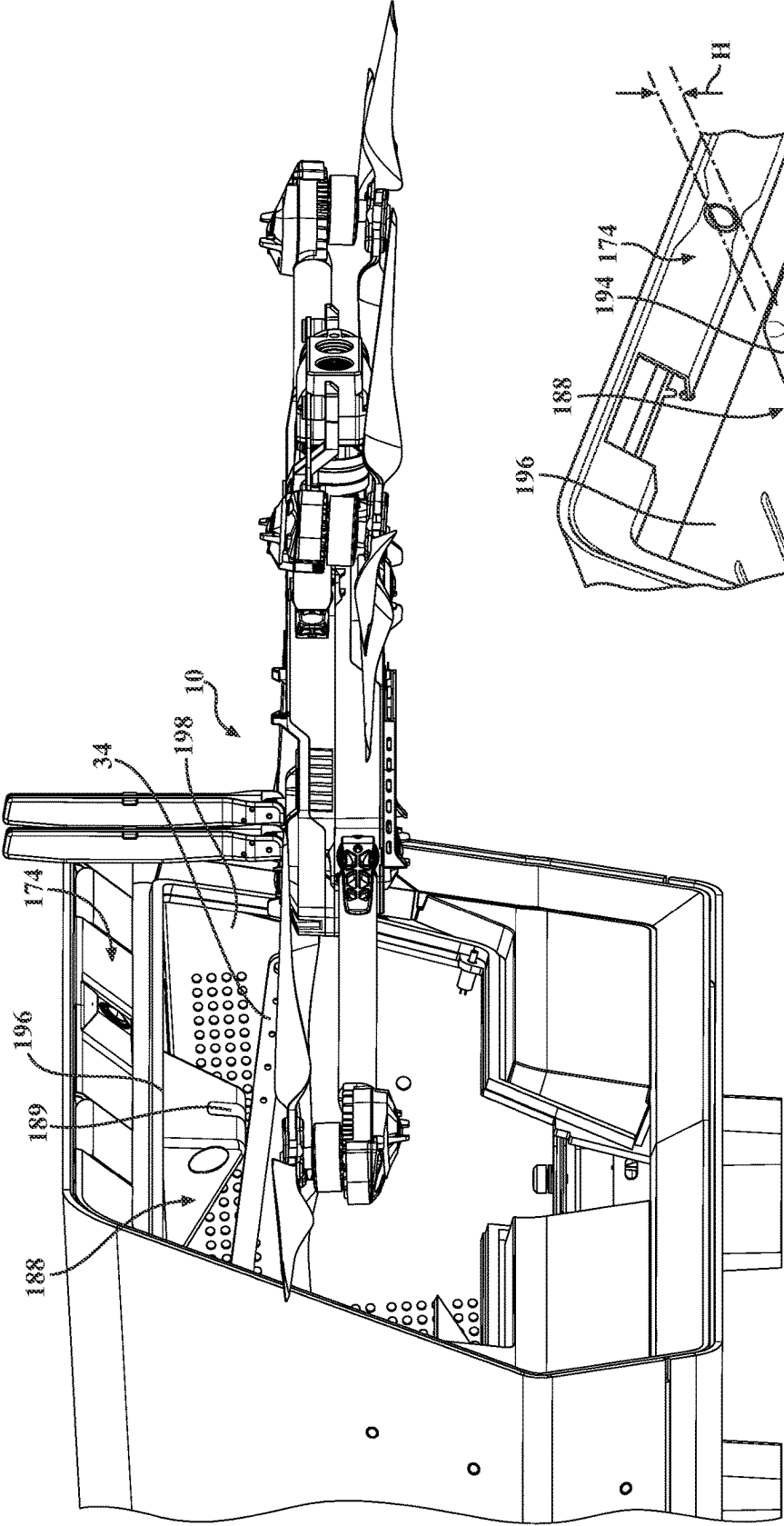
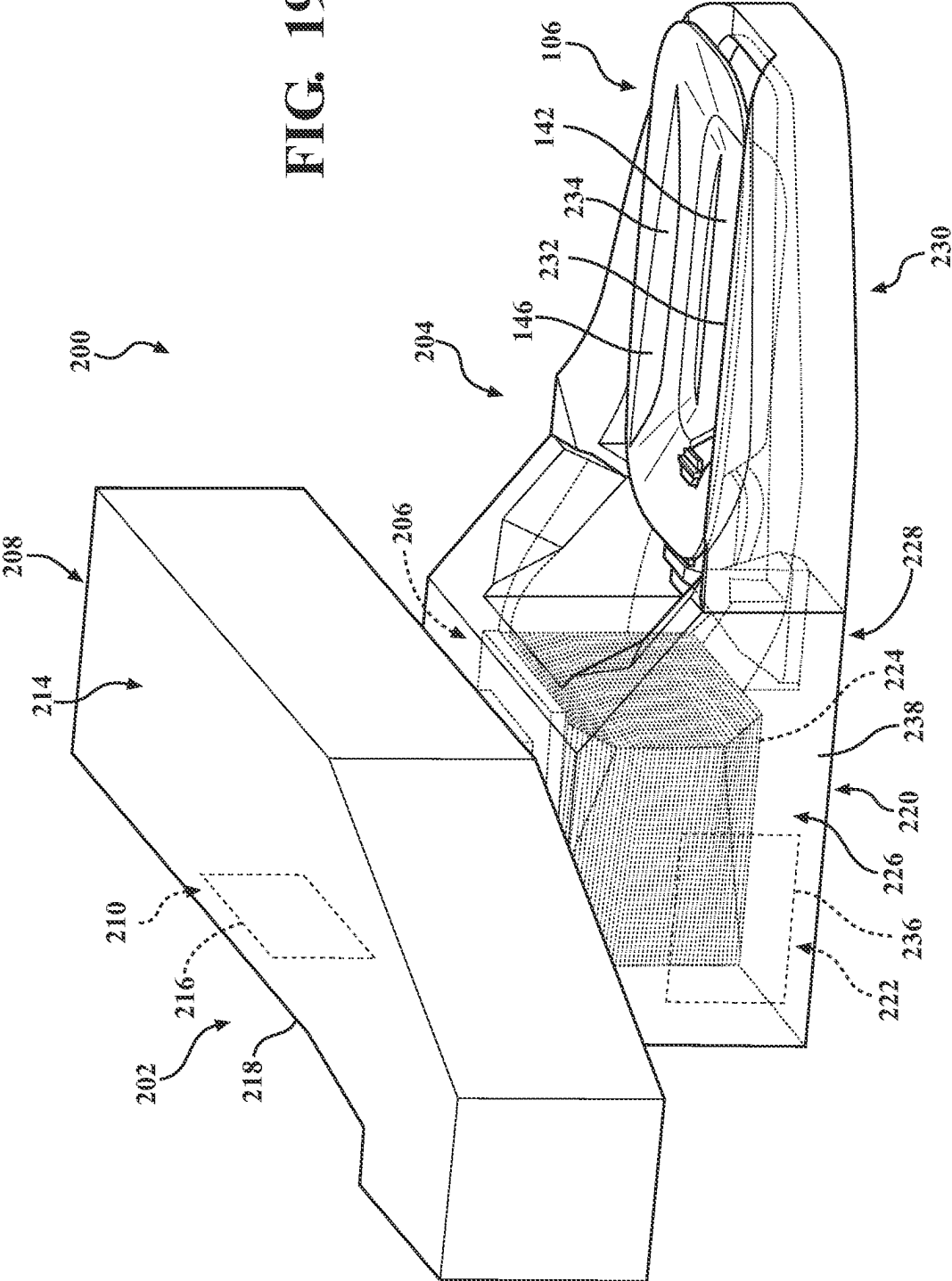


FIG. 18A

FIG. 18B

FIG. 19



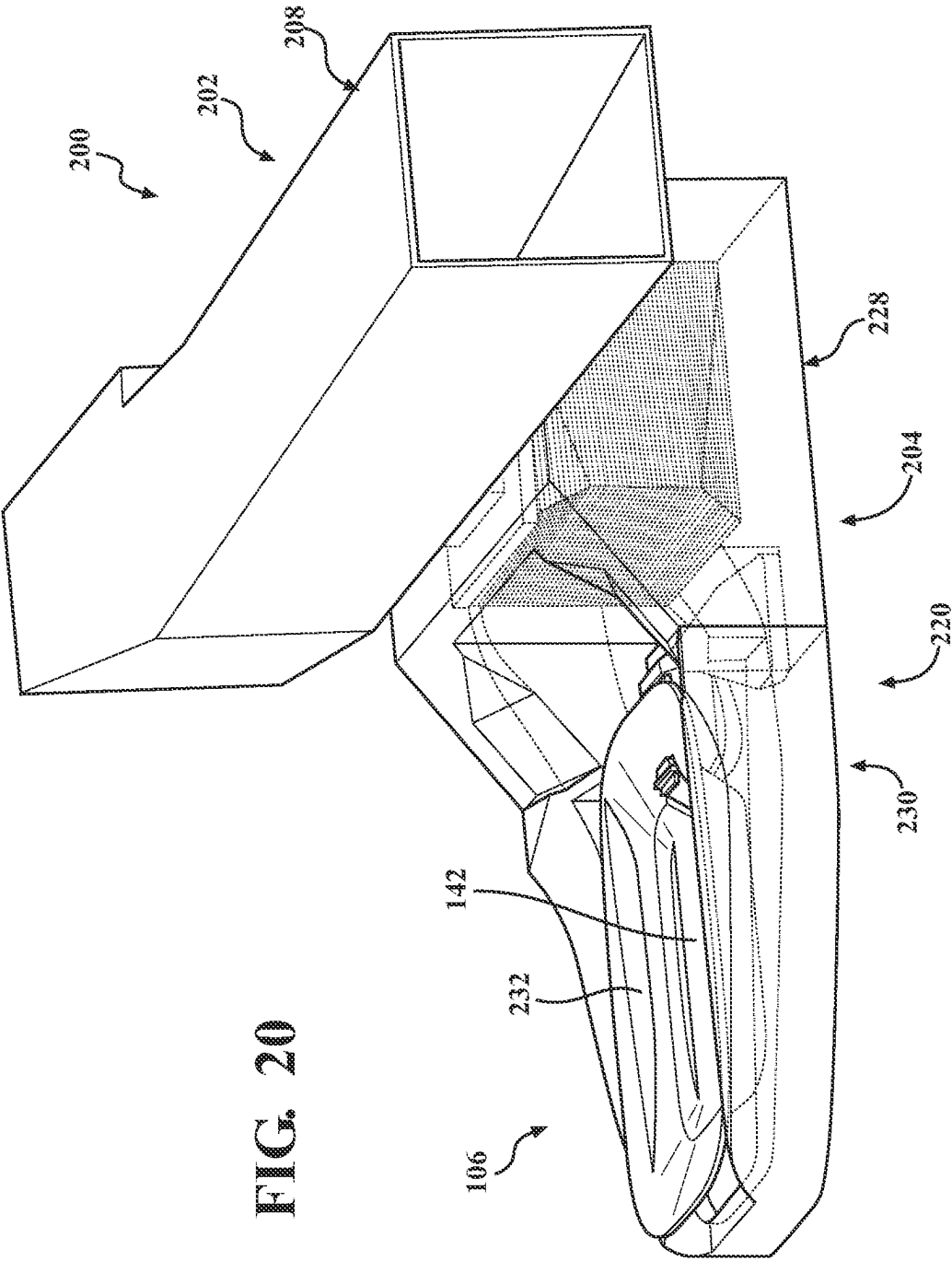


FIG. 21

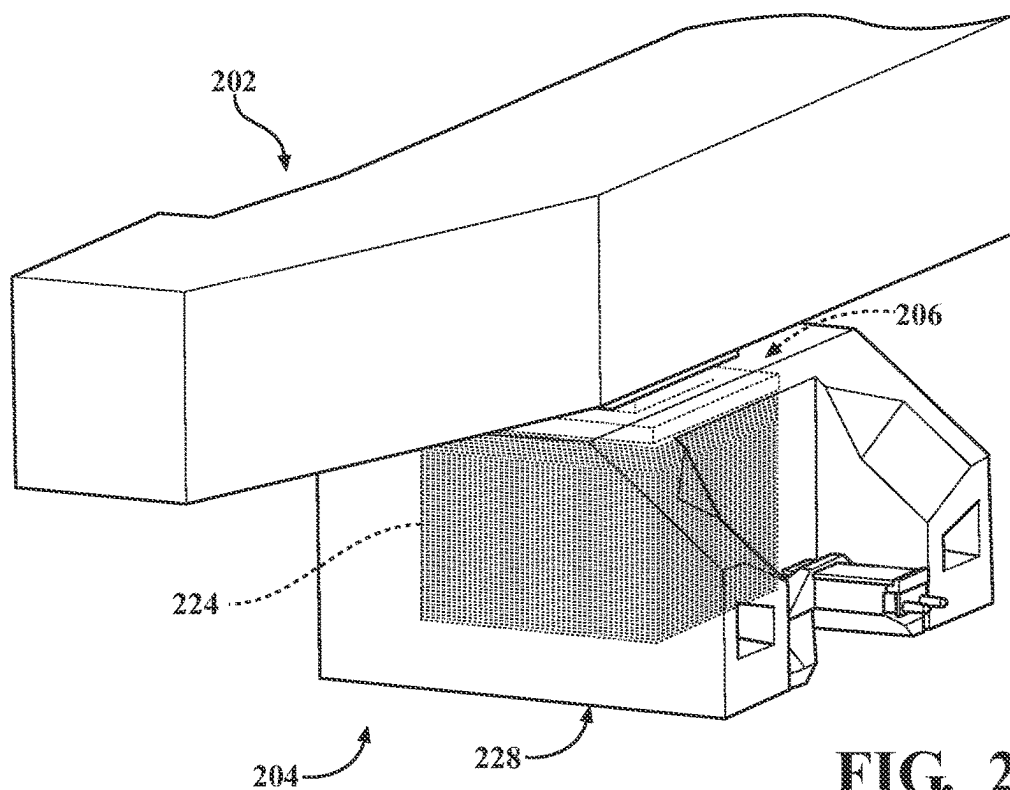
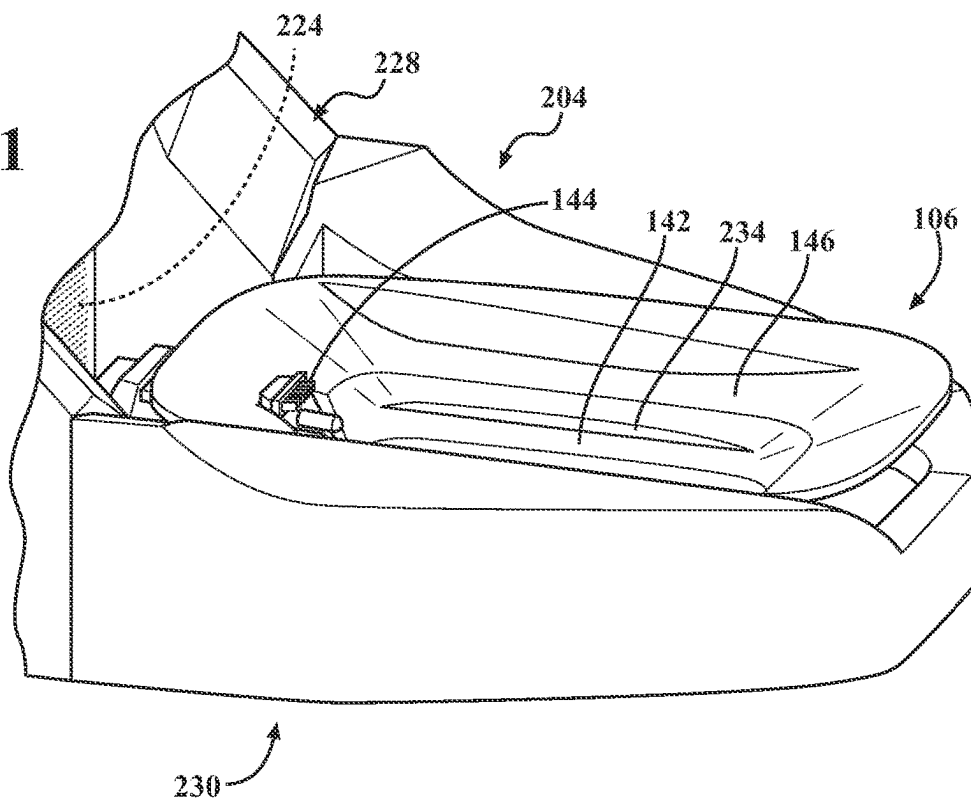


FIG. 22

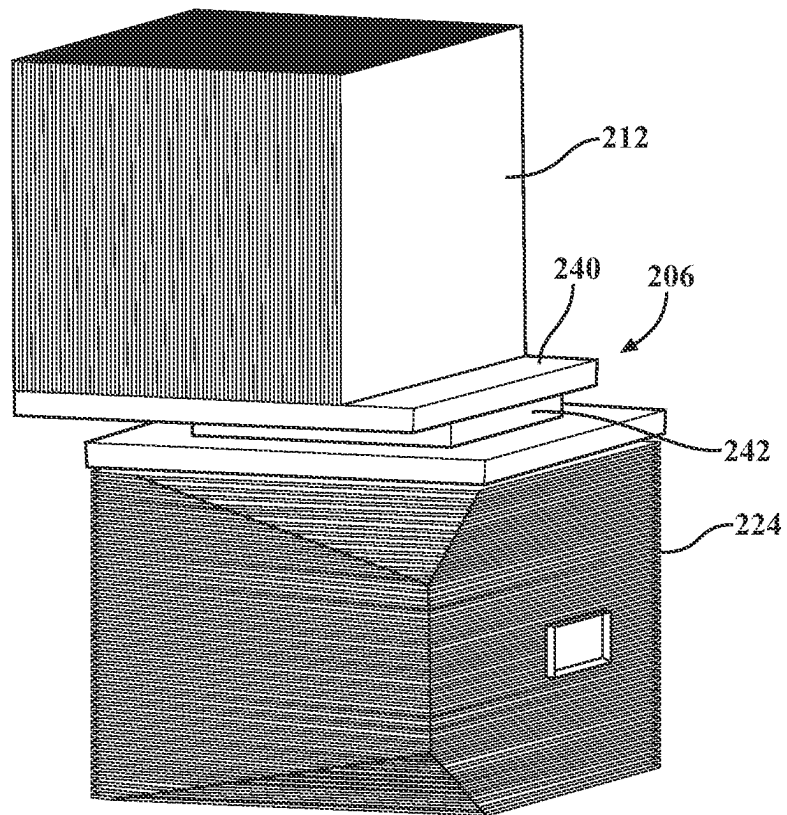


FIG. 23

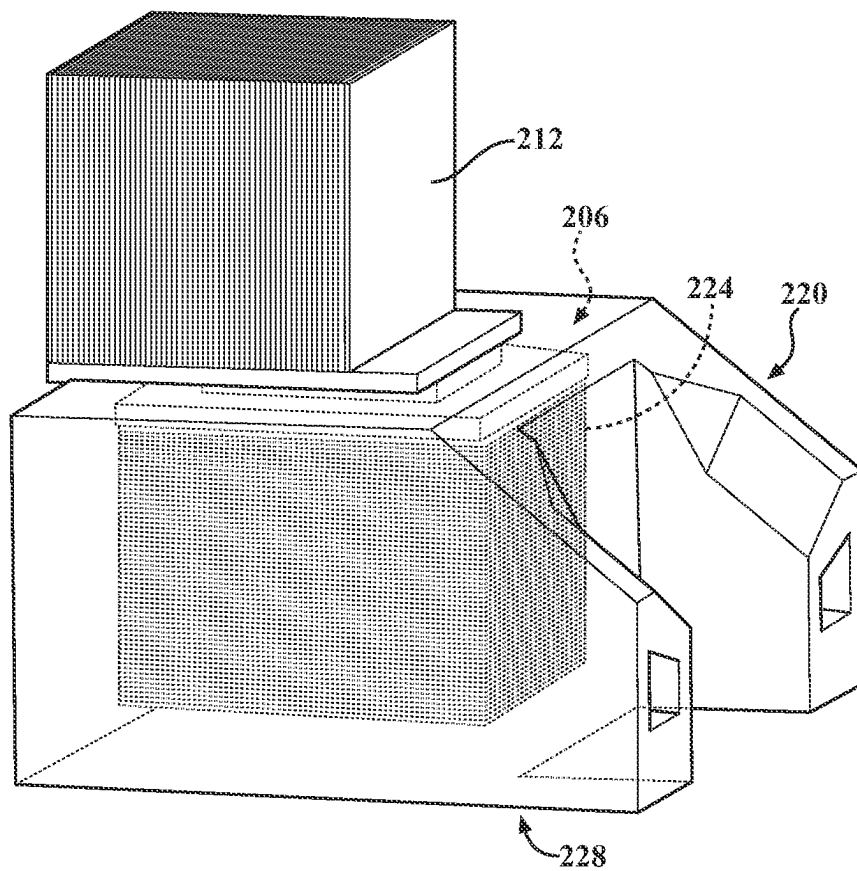


FIG. 24

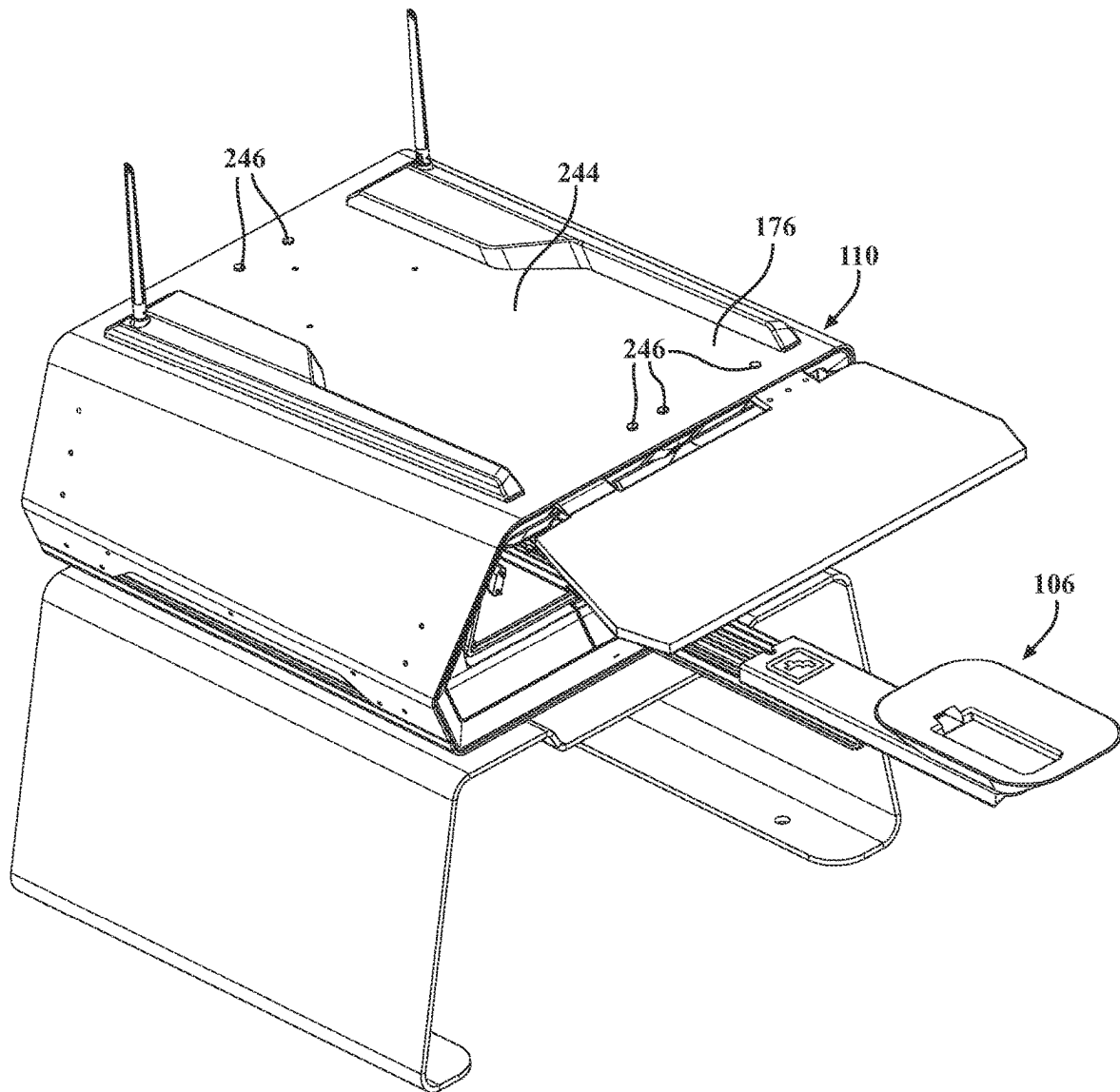


FIG. 25

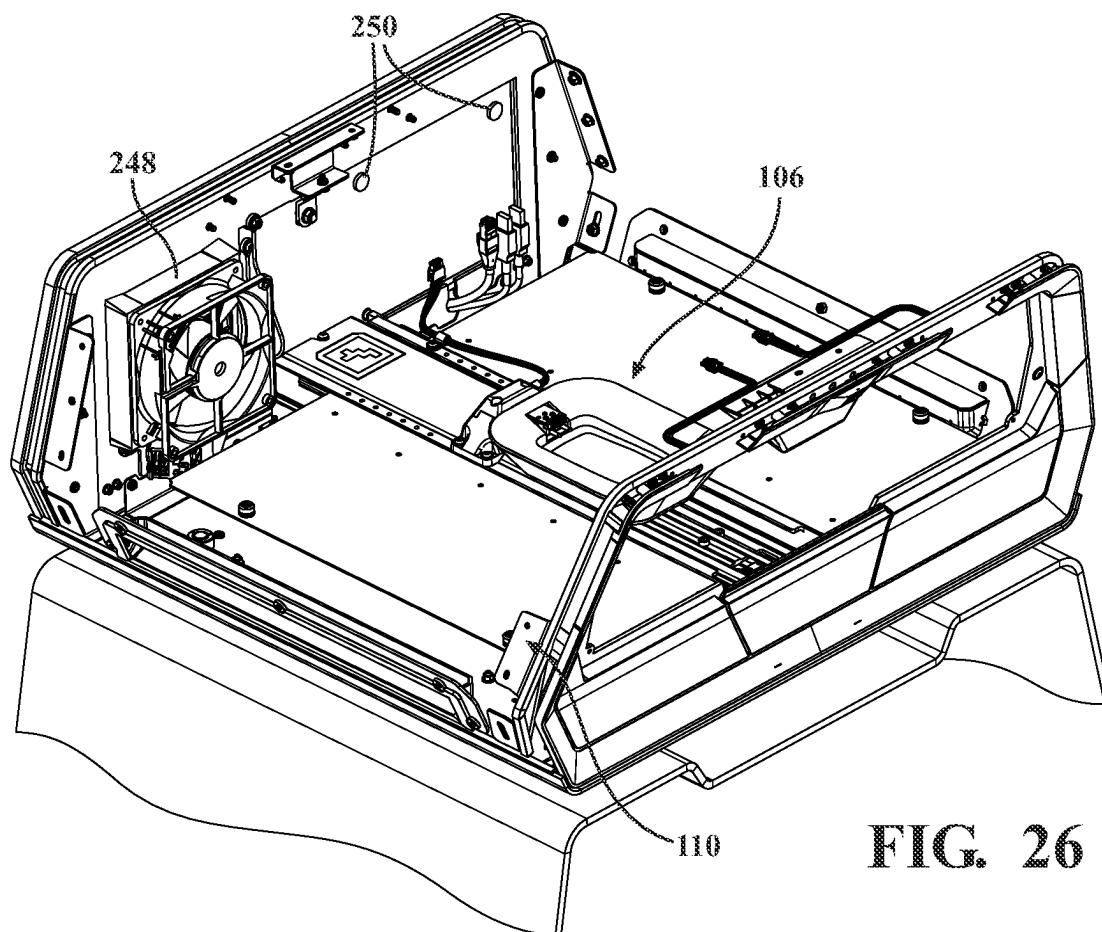


FIG. 26

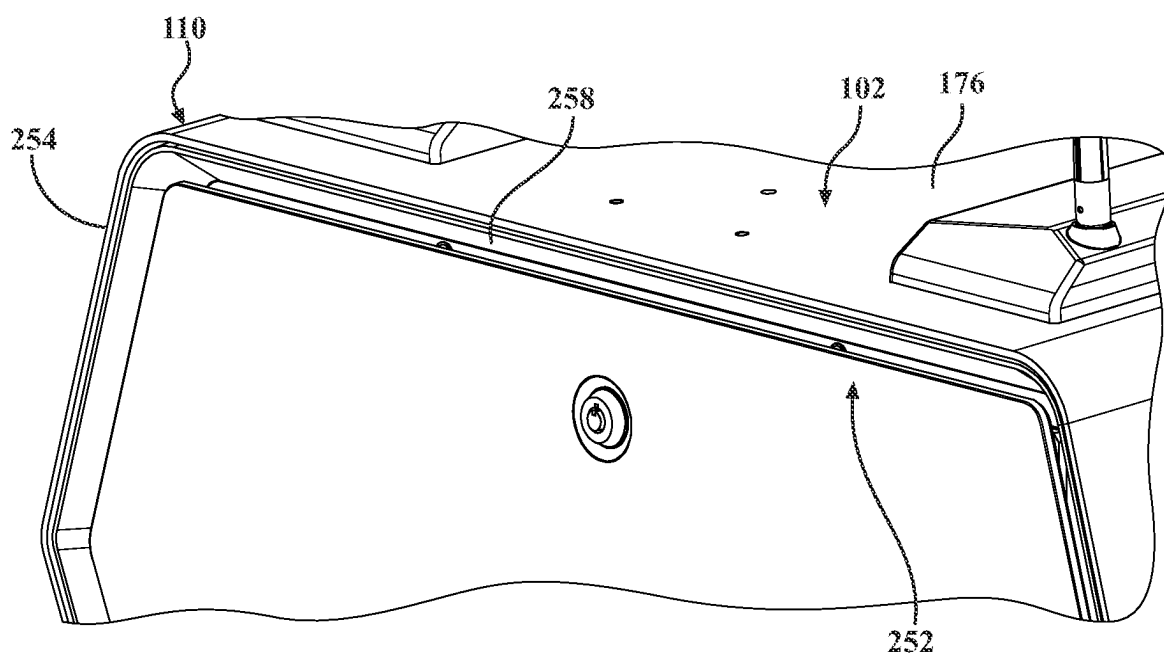


FIG. 27

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BASE STATIONS INCLUDING INTEGRATED SYSTEMS FOR SERVICING UAVS

CROSS-REFERENCE TO RELATED APPLICATION(S)

This application claims the benefit of and priority to U.S. Provisional Application No. 63/294,148, filed Dec. 28, 2021, U.S. Provisional Application No. 63/255,566, filed Oct. 14, 2021, and U.S. Provisional Application No. 63/222,768, filed Jul. 16, 2021, the disclosures of which are hereby incorporated by reference in their entirety.

TECHNICAL FIELD

The present disclosure relates to a base station (dock) for an unmanned aerial vehicle (UAV) (e.g., a drone). More specifically, the present disclosure relates to a base station that includes a series of integrated systems, which allow for automated servicing (e.g., docking, storage, charging, operation, etc.) and accommodation of a UAV.

BACKGROUND

In an attempt to manage various environmental settings and/or scenarios (e.g., temperature, precipitation, humidity, etc.), known base stations typically include a series of conditioning systems, which mandates the use of an enclosure that often drastically exceeds the size of the UAV(s) being serviced and/or accommodated. Additionally, conventional docking procedures often require the connection of a UAV to an external power supply during recharging or exchange of the UAV's power source itself. As a result, known base stations are often large, mechanically complex, and expensive.

The present disclosure addresses these deficiencies, among others, and provides a base station that offers improved servicing of UAVs and significant size reductions that allow for more efficient operation and substantial cost savings.

SUMMARY

In one aspect of the present disclosure, a base station is disclosed that is configured for use with an unmanned aerial vehicle (UAV). The base station includes: an enclosure; a cradle that is configured for electrical connection to a power source of the UAV during docking to facilitate charging of the power source; and a temperature control system that is connected to the cradle and which is configured to vary temperature of the power source of the UAV. The cradle is movable between a retracted position, in which the cradle is positioned within the enclosure, and an extended position, in which the cradle is positioned externally of the enclosure to facilitate docking with the UAV. The temperature control system includes: a thermoelectric conditioner (TEC) having a first end and a second end; a first air circuit that is thermally connected to the TEC and which is configured to regulate temperature of the TEC; and a second air circuit that is thermally connected to the TEC such that the TEC is located between the first air circuit and the second air circuit. The second air circuit is configured to direct air across the cradle to thereby heat or cool the power source of the UAV when docked with the base station.

In various embodiments, the temperature control system may be configured to cool the power source of the UAV (e.g., when the base station and the UAV are used in warmer

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environments) or to heat the power source of the UAV (e.g., when the base station and the UAV are used in cooler environments).

In certain embodiments, the first air circuit may be configured as an open system and the second air circuit may be configured as a closed system.

In certain embodiments, the TEC may be configured as a Peltier system.

In certain embodiments, the first air circuit may include: a first plenum; a first heat sink that is connected to the first plenum and the first end of the TEC; and a first air circulator that is configured to direct air through the first plenum and across the first heat sink to vary air temperature within the first air circuit and thereby regulate the temperature of the TEC.

In certain embodiments, the second air circuit may include: a second plenum; a second heat sink that is connected to the second plenum and the second end of the TEC; and a second air circulator that is configured to direct air through the second plenum and across the second heat sink to vary air temperature within the second air circuit and thereby heat or cool the power source of the UAV when docked with the base station.

In certain embodiments, the temperature control system may be configured to cool the power source of the UAV when docked with the base station.

In certain embodiments, the second plenum may define an air inlet and an air outlet.

In certain embodiments, the air inlet may be configured to direct air into the cradle and across the power source of the UAV and the air outlet may be configured to receive the air directed across the power source of the UAV and redirect the air across the second heat sink.

In certain embodiments, the second plenum may include a first section and a second section that is movable in relation to the first section.

In certain embodiments, the first section may be connected to the TEC and the second section may be connected to the cradle.

In certain embodiments, the first section and the second section may be configured for mating engagement upon movement of the cradle into the retracted position.

In another aspect of the present disclosure, a base station is disclosed that is configured for use with a UAV. The base station includes a temperature control system that is configured to vary temperature of the UAV. The temperature control system includes: a thermoelectric conditioner (TEC); an open air circuit that is thermally connected to the TEC and which is configured to regulate temperature of the TEC; and a closed air circuit that is thermally connected to the TEC such that the TEC is located between the open air circuit and the closed air circuit. The closed air circuit is configured to direct air across the UAV when docked with the base station.

In certain embodiments, the temperature control system may be configured to heat or cool the UAV subject to environmental conditions.

In certain embodiments, the closed air circuit may include a first section and a second section that is movable in relation to the first section.

In certain embodiments, the base station may further include a cradle that is configured for electrical connection to the UAV during docking to facilitate charging of the UAV.

In certain embodiments, the cradle may be extendable from and retractable into the base station.

In certain embodiments, the first section of the closed air circuit may be connected to the TEC and the second section of the closed air circuit may be connected to the cradle.

In certain embodiments, the first section and the second section may be configured for mating engagement upon retraction of the cradle into the base station.

In another aspect of the present disclosure, a method is disclosed for regulating the temperature of a power source in a UAV. The method includes: docking the UAV within a cradle of a base station; retracting the cradle into the base station; and directing thermally conditioned air across the power source of the UAV via an air circuit that is connected to the cradle.

In certain embodiments, the method may further include directing air across a heat sink that is thermally connected to a thermoelectric conditioner (TEC) to treat the air prior to direction across the power source of the UAV.

In certain embodiments, directing air across the heat sink may include circulating the air through a plenum that is connected to the heat sink.

In certain embodiments, retracting the cradle into the base station may include closing the air circuit.

In certain embodiments, closing the air circuit may include mating a first section of the plenum with a second section of the plenum.

In certain embodiments, the first section of the plenum may be connected to the TEC and the second section of the plenum may be connected to the cradle.

In another aspect of the present disclosure, a base station is disclosed that is configured for use with a UAV. The base station includes: an enclosure with an outer housing that defines a roof section and an inner housing that is connected to the outer housing; one or more heating elements that are supported by the enclosure and which are configured to heat the roof section; one or more fiducials that are supported by the enclosure; an illumination system that is supported by the enclosure and which is configured to illuminate the one or more fiducials; and a visualization system that is supported by the enclosure.

In certain embodiments, the enclosure (e.g., the outer housing) may define one or more channels that are configured to direct water in a manner that inhibits entry into the base station.

In certain embodiments, the base station may further include one or more temperature sensors that are in communication with the one or more heating elements such that the one or more heating elements are activated upon receiving a signal relayed by the one or more temperature sensors indicating that temperature has crossed a threshold.

In certain embodiments, the one or more fiducials may include a first fiducial and a second fiducial, each of which is supported by the roof section.

In certain embodiments, the second fiducial may be removably connected to the roof section.

In certain embodiments, the first fiducial may define a first surface area, and the second fiducial may define a second surface area that is less than the first surface area.

In certain embodiments, the first surface area may lie substantially within the range of (approximately) 40 percent to (approximately) 80 percent of a surface area defined by the roof section.

In certain embodiments, the second surface area may lie substantially within the range of (approximately) 10 percent to (approximately) 50 percent of the first surface area.

In certain embodiments, the illumination system may include one or more light sources that are secured to the roof section and which are configured to light the first fiducial and the second fiducial.

In certain embodiments, the illumination system may be configured to strobe the one or more light sources according to a pattern that is recognizable by the UAV during approach to thereby identify the base station.

In certain embodiments, the visualization system may include a digital image capturing device that is configured to identify precipitation and actuate the one or more heating elements.

In certain embodiments, the base station may further include one or more status indicators that are supported by the enclosure (e.g., the outer housing).

In certain embodiments, the base station may further include one or more internal fans to regulate temperature and/or humidity within the base station.

In certain embodiments, the internal fan(s) may be supported by at least one of the outer housing and the inner housing.

In another aspect of the present disclosure, a base station is disclosed that is configured for use with a UAV. The base station includes: an enclosure; a door that is movably connected to the enclosure such that the door is repositionable between a closed position and an open position; and one or more actuators that extend between the door and the enclosure. The enclosure includes an outer housing and an inner housing that is connected to the outer housing and which defines an internal cavity that is configured receive the UAV. Each actuator includes a motor assembly and a linkage assembly that extends between the motor assembly and the door. The motor assembly is secured to the inner housing such that the motor assembly is located between the outer housing and the inner housing, and the linkage assembly extends through the inner housing.

In certain embodiments, the linkage assembly may include: a drive screw that is operatively connected to the motor assembly such that actuation of the motor assembly causes rotation of the drive screw; a carrier that is threadably engaged to the drive screw such that rotation of the drive screw causes axial translation of the carrier; a first arm; and a second arm.

In certain embodiments, the first arm may have a first end that is pivotably connected to the carrier and a second end, and the second arm may have a first end that is pivotably connected to the second end of the first arm and a second end that is pivotably connected to the door.

In certain embodiments, the base station may further include a bracket that is fixedly connected to the door.

In certain embodiments, the bracket may be pivotably connected to the second end of the second arm.

In certain embodiments, the drive screw may be configured such that rotation of the drive screw in a first direction causes advancement of the carrier towards the door and rotation of the drive screw in a second direction causes advancement of the carrier away from the door.

In certain embodiments, the drive screw may include threading defining a pitch that is configured to inhibit force transmission from the door to the carrier to thereby maintain the door in the closed position.

In another aspect of the present disclosure, a base station is disclosed that is configured for use with an unmanned aerial vehicle (UAV). The base station includes: an enclosure defining an internal cavity that is configured to receive the UAV; a first fiducial that is supported by a roof section of the enclosure and which defines a first surface area; a

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second fiducial that is supported by the roof section of the enclosure and which defines a second surface area that is less than the first surface area; and an illumination system that is supported by roof section and which is configured to illuminate the first fiducial and the second fiducial.

In certain embodiments, the second fiducial may be configured for removable connection to the roof section.

In certain embodiments, the first surface area may lie substantially within the range of (approximately) 40 percent to (approximately) 80 percent of a surface area defined by the roof section.

In certain embodiments, the second surface area may lie substantially within the range of (approximately) 10 percent to (approximately) 50 percent of the first surface area.

In certain embodiments, the illumination system may be configured to strobe according to a pattern recognizable by the UAV during approach to thereby identify the base station.

In another aspect of the present disclosure, a UAV is disclosed that includes a power source. The power source includes: one or more power cells; one or more thermal transfer members that are thermally connected to the one or more power cells; and a heat exchanger that is thermally connected to the one or more thermal transfer members such that the one or more thermal transfer members and the heat exchanger facilitate a transfer of thermal energy between the power source and ambient air to decrease or increase temperature of the power source.

In certain embodiments, the one or more thermal transfer members may extend between the one or more power cells and the heat exchanger.

In certain embodiments, the one or more thermal transfer members may include graphite.

In certain embodiments, the one or more thermal transfer members may be unitary in construction.

In certain embodiments, the one or more power cells and the one or more thermal transfer members may correspond in number.

In certain embodiments, the one or more power cells may include a plurality of individual power cells and the one or more thermal transfer members may include a plurality of individual thermal transfer members.

In certain embodiments, the heat exchanger may include one or more diffusers to increase surface area of the heat exchanger and thermal energy distribution towards or away from the power source.

In certain embodiments, the one or more diffusers may extend axially and/or laterally along an outer surface of the heat exchanger.

In certain embodiments, the one or more diffusers may include a plurality of diffusers.

In certain embodiments, the plurality of diffusers may be configured as fins that define one or more channels therebetween.

In certain embodiments, the one or more channels may be configured to direct air flow along the heat exchanger to increase thermal energy distribution towards or away from the power source.

In another aspect of the present disclosure, a UAV is disclosed that includes a power source. The power source includes a heat exchanger that is configured to transfer thermal energy between the power source and ambient air to decrease or increase temperature of the power source. The heat exchanger includes one or more diffusers to increase surface area of the heat exchanger and thermal energy distribution towards or away from the power source.

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In certain embodiments, the one or more diffusers may extend axially and/or laterally along an outer surface of the heat exchanger.

In certain embodiments, the one or more diffusers may include a plurality of diffusers.

In certain embodiments, the plurality of diffusers may be configured as fins that define one or more channels therebetween.

In certain embodiments, the one or more channels may be configured to direct air flow along the heat exchanger to increase thermal energy distribution towards or away from the power source.

In another aspect of the present disclosure, a UAV is disclosed that includes: a body; one or more antennas that extend from the body; and a power source that is connected to the body. The one or more antennas are reconfigurable between an active configuration, in which the one or more antennas extend outwardly from the body, and a passive configuration, in which the one or more antennas are positioned adjacent to the body. The power source includes one or more diffusers that extend axially and/or laterally along an outer surface of the power source to transfer thermal energy between the power source and ambient air to decrease or increase temperature of the power source.

In certain embodiments, the one or more antennas may be biased towards the active configuration.

In certain embodiments, the one or more diffusers may include a plurality of diffusers.

In certain embodiments, the plurality of diffusers may be configured as fins that define one or more channels therebetween.

In certain embodiments, the one or more channels may be configured to direct air flow along the power source to increase thermal energy distribution towards or away from the power source.

In certain embodiments, the power source may further include one or more power cells and one or more thermal transfer members that are thermally connected to the one or more power cells and to the one or more diffusers.

In certain embodiments, the one or more thermal transfer members may include graphite.

In certain embodiments, the one or more thermal transfer members may be unitary in construction.

BRIEF DESCRIPTION OF THE DRAWINGS

The present disclosure is best understood from the following detailed description when read in conjunction with the accompanying drawings. It is emphasized that, according to common practice, the various features of the drawings are not to-scale. On the contrary, the dimensions of the various features are arbitrarily expanded or reduced for clarity.

FIG. 1 is a front, perspective view of a base station (dock) according to the principles of the present disclosure.

FIG. 2 is a partial, side view of the base station shown with one embodiment of a UAV docked within a cradle of the base station.

FIG. 3 is a partial, side view of the base station shown with an alternate embodiment of the UAV seen in FIG. 2.

FIG. 4 is a bottom, perspective view of a power source of the UAV.

FIG. 5 is a partial, side, perspective view of the power source of the UAV.

FIG. 6 is a side, schematic view of the base station.

FIG. 7 is a partial, rear, perspective view of the base station.

FIG. 8 is a partial, side, perspective view of the base station illustrating a plurality of actuators that are configured to open and close a door of the base station.

FIGS. 9 and 10 are partial, top, perspective views of the base station illustrating the plurality of actuators.

FIG. 11 is a partial, front, perspective view of the base station illustrating a cap that is supported on a slide mechanism, which facilitates movement of the cradle between a retracted position and an extended position.

FIG. 12 is a partial, rear, perspective view of the cap shown overlaid by the door of the base station upon closure.

FIG. 13 is a partial, side, perspective view of the base station with the cradle shown in the extended position.

FIG. 14 is a partial, top, perspective view of the base station illustrating one or more connection antennas and one or more fiducials that are supported on a roof section thereof.

FIG. 15 is a partial, top, perspective view of the base station according to an embodiment of the disclosure that includes a plurality of fiducials on the roof section.

FIG. 16 is a partial, front, perspective view of the base station illustrating a status indicator.

FIG. 17 is a partial, top, perspective view of the base station according to an embodiment of the disclosure that includes an illumination system on the roof section.

FIG. 18A is a partial, front, perspective view of the base station during movement of the cradle into the retracted position after docking of the UAV.

FIG. 18B is a partial, front, perspective view of the base station during movement of the cradle into the retracted position after docking of the UAV illustrating one embodiment of a contact member that is configured to facilitate reconfiguration (e.g., folding) of one or more antennas on the UAV.

FIGS. 19 and 20 are opposite side, schematic views of a temperature control system of the base station, which includes a first air circuit; a second air circuit; and a thermoelectric conditioner (TEC).

FIG. 21 is a partial, side, schematic view of the cradle and the second air circuit.

FIG. 22 is partial, side, schematic view of the temperature control system.

FIG. 23 is a partial, side, schematic view of the temperature control system illustrating a first heat sink and a second heat sink.

FIG. 24 is a partial, side, schematic view of the first heat sink and the second heat sink, which is shown located within a plenum of the second air circuit.

FIG. 25 is a partial, top, schematic view of the base station according to an embodiment of the disclosure that includes one or more heating elements on the roof section.

FIG. 26 is a partial, top, schematic view of the base station illustrating an internal fan.

FIG. 27 is a partial, rear, perspective view of the base station illustrating one or more channels that collect and direct water.

DETAILED DESCRIPTION

Overview

The present disclosure relates to a base station for use with a UAV that is configured to not only charge the power source of the UAV, but regulate the temperature of the power source of the UAV. In various embodiments of the disclosure, depending upon the environmental conditions, the base station may be configured to cool the power source of the UAV (e.g., when the base station and the UAV are used in

warmer environments) or heat the power source of the UAV (e.g., when the base station and the UAV are used in cooler environments).

To facilitate cooling and/or heating of the power source of the UAV, the base station includes a temperature control system. The temperature control system includes: a thermoelectric conditioner (TEC); a first air circuit that is thermally connected to the TEC and which is configured to regulate (e.g., increase or decrease) temperature of the TEC; and a second air circuit that is thermally connected to the TEC such that the TEC is located between the first air circuit and the second air circuit. The second air circuit is configured to direct treated air (e.g., air that has been either cooled or heated) across the power source of the UAV to thereby heat or cool the power source (subject to environmental conditions).

To increase functionality and improve operation of the base station, the base station includes a plurality of additional (ancillary) systems that are configured to address environmental concerns (e.g., humidity, precipitation, etc.), security concerns (e.g., anti-theft systems and mechanisms), etc. For example, the base station may include: a heating element that is supported by a roof section to reduce the presence of snow and/or ice; one or more fiducials that facilitate visual identification of the base station by the UAV; an illumination system that improves visibility of the one or more fiducials (e.g., during nighttime operation); and a visualization system (e.g., a digital image capturing device) that supports observation and visual analysis of the environment in which the base station and the UAV are located.

Referring now to the drawings, FIGS. 1 and 2 illustrate an unmanned aerial vehicle (UAV) 10 and a base (docking) station 100 that is configured for automated servicing (e.g., docking, storage, charging, operation, etc.) and accommodation of the UAV 10. While a single UAV 10 and a single base station 100 are shown and described herein, in certain embodiments of the disclosure, it is envisioned that a plurality of UAVs 10 and a plurality of base stations 100 may be utilized depending, for example, upon the particular intended use of the UAVs 10.

The UAV

The UAV 10 includes one or more propulsion mechanisms (systems) 12 and a power source 14 (e.g., a battery 16). To support autonomous landing and docking of the UAV 10 with the base station 100, it is envisioned that the UAV 10 may follow any suitable process or procedure and may include any suitable electrical and/or logic components, as described in U.S. application Ser. No. 16/991,122 ("the '122 application"), the entire contents of which are hereby incorporated by reference.

The propulsion mechanism(s) 12 may include any components and/or structures suitable for the intended purpose of supporting flight of the UAV 10. For example, as seen in FIG. 2, it is envisioned that the propulsion mechanism(s) 12 may include propellers 18. It should be appreciated, however, that the particular configuration and/or components of the propulsion mechanism(s) 12 may be varied without departing from the scope of the present disclosure. FIG. 3, for example, illustrates an alternate configuration for the UAV 10.

Although shown as including four propulsion mechanisms 12 in the particular embodiment of the UAV 10 described herein, it should be appreciated that the particular number of propulsion mechanisms 12 may be varied without departing from the scope of the present disclosure. As such, embodiments of the UAV 10 including fewer and greater

numbers of propulsion mechanisms **12** are also envisioned herein and are not beyond the scope of the present disclosure.

It is envisioned that the propulsion mechanism(s) **12** may include either a fixed configuration or a variable configuration. For example, it is envisioned that the propulsion mechanism(s) **12** may be reconfigurable between an extended (first) configuration and a collapsed (folded, second) configuration to allow for a reduction in the overall size of the UAV **10** (e.g., during entry into the base station **100**) and, thus, a reduction in the overall size of the base station **100**, as described in further detail below.

The power source **14** is located (e.g., attached to or otherwise supported on) a lower (bottom) surface of the UAV **10** and includes one or more conducting (electrical) contacts (not shown) that are configured for engagement (contact) with one or more corresponding conducting (electrical) contacts on the base station **100** to enable charging of the power source **14**.

Heat Exchange

As seen in FIGS. **4** and **5**, the power source **14** includes one or more power cells **20**; one or more thermal transfer members **22**; and a heat exchanger **24**, which defines a lower (bottom) surface of the power source **14** and acts as a cover for the power cell(s) **20** and the thermal transfer member(s) **22**. The thermal transfer member(s) **22** and the heat exchanger **24** are configured to facilitate the transfer of thermal energy between the power source **14** and the ambient (air) to decrease or increase temperature of the power source **14**. For example, depending upon the particular environment in which the UAV **10** and the base station **100** are employed, the thermal transfer member(s) **22** and the heat exchanger **24** may be utilized to dissipate heat generated by the power source **14** during use and/or charging of the UAV **10** (e.g., in warmer environments) or to transfer heat to the power source **14** (e.g., in cooler environments) to thereby improve efficiency, operation, and/or the usable life of the UAV **10** and/or the power source **14**.

In the particular embodiment illustrated, the power source **14** includes a plurality of individual (e.g., discrete) power cells **20**. It should be appreciated, however, that the particular number and/or configuration of the power cells **20** may be varied in alternate embodiments without departing from the scope of the present disclosure. For example, embodiments of the power source **14** including a single power cell **20** are also envisioned herein.

The thermal transfer member(s) **22** are thermally connected to, and extend between, the power cell(s) **20** and the heat exchanger **24**. In the particular embodiment of the disclosure illustrated, the power source **14** includes a plurality of individual (e.g., discrete) thermal transfer members **22**, each of which is associated with (e.g., thermally connected to) a corresponding power cell **20** (e.g., such that the power source **14** includes a corresponding (equal) number of power cells **20** and thermal transfer members **22**). It should be appreciated, however, that the particular number of the thermal transfer members **22** may be varied in alternate embodiments without departing from the scope of the present disclosure and that embodiments of the power source **14** including an unequal number of power cells **20** and thermal transfer members **22** are also envisioned herein. For example, the present disclosure contemplates embodiments in which the number of power cells **20** may exceed the number of thermal transfer members **22** are also envisioned herein (e.g., embodiments in which the power source **14** includes a single thermal transfer member **22** that extends between the collection of power cells **20** and the heat

exchanger **24**) as well as embodiments in which the number of thermal transfer members **22** may exceed the number of power cells **20**.

The thermal transfer member(s) **22** may include (e.g., may be formed partially or entirely from) any material or combination of materials that is suitable for the intended purpose of transferring heat between, and thermally connecting, the power cell(s) **20** and the heat exchanger **24**. For example, in one particular embodiment, it is envisioned that the thermal transfer member(s) **22** may include (e.g., may be formed partially or entirely from) graphite. It should be appreciated, however, that the use of other materials would not be beyond the scope of the present disclosure. Additionally, although each thermal transfer member **22** is shown as being unitary in construction (i.e., as being formed from a single piece of material), in alternate embodiments of the disclosure, it is envisioned that each thermal transfer member **22** may include a series of individual segments that are connected to each other during manufacture, assembly of the power source **14**, or at any other suitable point in time.

The heat exchanger **24** is thermally connected to the thermal transfer member(s) **22** and is configured to communicate and distribute thermal energy between the power source **14** and the ambient (air), either away from the power source **14** (e.g., when utilized in warmer environments) or towards the power source **14** (e.g., when utilized in cooler environments), and may include (e.g., may be formed partially or entirely from) any material or combination of materials suitable for that intended purpose. For example, it is envisioned that the heat exchanger **24** may include (e.g., may be formed partially or entirely from) aluminum, magnesium, copper, etc.

To increase the available surface area and, thus, the distribution of thermal energy (either towards or away from the power source **14**), in certain embodiments, such as that illustrated throughout the figures, the heat exchanger **24** may include one or more diffusers **26**, which may be configured in any manner suitable for that intended purpose. For example, it is envisioned that the diffuser(s) **26** may be configured as pins, protrusions, ribs, or other such surface irregularities and may extend axially (e.g., along a longitudinal axis Y) and/or laterally (e.g., along a transverse axis X) along an outer (bottom) surface **28** of the heat exchanger **24**. In the particular embodiment of the heat exchanger **24** illustrated throughout the figures, for example, the diffusers **26** are configured as fins **30** that define a plurality of channels **32** therebetween, which collectively direct air flow along the heat exchanger **24** to further increase the distribution of thermal energy.

Although shown as including a plurality of diffusers **26** and channels **32** in the particular embodiment of the disclosure illustrated, it should be appreciated that the particular number of the diffusers **26** and/or channels **32** may be varied in alternate embodiments without departing from the scope of the present disclosure. For example, embodiments of the heat exchanger **24** including a single diffuser **26** are also envisioned herein.

Base Station Construction

With reference now to FIGS. **4-13** as well, the base station **100** includes: an enclosure (body) **102**; a door **104** that is movably connected to the enclosure **102**; and a cradle **106** that is configured to receive (accommodate) the UAV **10**.

The enclosure **102** includes an inner housing (shell) **108** and an outer housing (cover) **110**. The respective inner and outer housings **108**, **110** are configured as separate, discrete structures that may be connected together in any suitable manner, whether fixedly or removably (e.g., to allow for

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repeated assembly and disassembly of the base station 100 during maintenance, repair, etc.). For example, it is envisioned that the respective inner and outer housings 108, 110 may be connected via a plurality of mechanical fasteners (e.g., screws, pins, bolts, clips, etc.), which may be hidden (or otherwise obscured) to inhibit theft and/or unauthorized disassembly of the base station 100.

The inner housing 108 defines an internal cavity 112 that is configured to receive and accommodate the UAV 10. Additionally, the inner housing 108 provides a mounting surface for various components of the base station 100 including, for example, electrical components, actuators, and the like, which are secured (mounted) to the inner housing 108 and support operation of the base station 100.

The outer housing 110 provides structural support to the base station 100 and protects the inner housing 108 and the various components that are secured (mounted) thereto (e.g., from dust, debris, the ingress of moisture and/or water, etc.). Additionally, the outer housing 110 supports various external components of the base station 100, as described in further detail below.

To facilitate access to the various components accommodated within the outer housing 110 (and/or the inner housing 108), in certain embodiments, it is envisioned that the outer housing 110 may include an access panel 114 (FIG. 7) to support maintenance, repair, etc., which may be incorporated in any suitable location (e.g., at a rear of the outer housing 110). To inhibit theft and/or unauthorized access, it is envisioned that the access panel 114 may include a locking mechanism or other such suitable safeguard.

In certain embodiments of the disclosure, it is envisioned that the outer housing 110 may include exterior coloration that not only reduces solar loading (heating), but promotes contrast to facilitate visualization and/or identification of the base station 100 by the UAV 10 during docking, as described in further detail below.

The door 104 is movably connected to the outer housing 110 such that the door 104 is repositionable between a closed position (FIG. 1), in which the door 104 conceals the internal cavity 112, and an open position (FIG. 6), in which the internal cavity 112 is exposed. More specifically, the door 104 is movably connected to a forward frame 116 of the outer housing 110. Although illustrated as being pivotably connected to the forward frame 116 in the particular embodiment illustrated throughout the figures, it is also envisioned that the door 104 may be slidably repositionable between the closed position and the open position in alternate embodiments of the disclosure.

To facilitate movement of the door 104 between the closed position and the open position, the base station 100 includes one or more actuators 118 (FIGS. 8-10) that extend between the door 104 and the enclosure 102. Although shown as including a plurality (e.g., two) actuators 118 in the particular embodiment illustrated, which are secured to opposing lateral ends of the door 104, it should be appreciated that the present disclosure also contemplates embodiments in which a single actuator 118 may be utilized to control the position of the door 104 (e.g., to reduce the overall cost and complexity of the base station 100).

Each actuator 118 includes a motor assembly 120 (e.g., a stepper motor) and a linkage assembly 122 that extends between the motor assembly 120 and the door 104. More specifically, each motor assembly 120 is secured to the inner housing 108 such that the motor assembly(ies) 120 are located between the inner housing 108 and the outer housing 110, which protects the motor assembly(ies) 120 and inhibits the collection of any dust, debris, etc. The linkage assembly

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122 extends from the motor assembly 120, through the inner housing 108, and pivotably engages the door 104 such that, upon actuation of the motor assembly 120, the linkage assembly 122 applies a force to the door 104 to thereby facilitate movement of the door 104 between the closed position and the open position.

Each linkage assembly 122 includes: a (threaded) drive screw 124; a carrier 126; a first arm 128; and a second arm 130. The drive screw 124 is (operatively) connected to the motor assembly 120 such that actuation of the motor assembly 120 causes rotation of the drive screw 124. The carrier 126 is threadably engaged to the drive screw 124 such that rotation of the drive screw 124 causes axial translation of the carrier 126. More specifically, rotation of the drive screw in a first direction (e.g., clockwise) causes forward advancement of the carrier 126 (e.g., movement of the carrier 126 towards the door 104) and rotation of the drive screw in a second direction (e.g., counterclockwise) causes rearward advancement of the carrier 126 (e.g., movement of the carrier 126 away from the door 104). The first arm 128 includes a first end 132 that is connected to the carrier 126 (either fixedly or pivotably) and a second end 134 that is pivotably connected to the second arm 130. The second arm 130 includes a first end 136 that is pivotably connected to the second end 134 of the first arm 128 and a second end 138 that is pivotably connected to a bracket 140. The bracket 140 is fixedly connected to the door 104 which allows for the transmission of force from the carrier 126 to the door 104 via the arms 128, 130.

The cradle 106 is configured for electrical connection to the power source 14 of the UAV 10 during docking to facilitate (support) charging of the power source 14. More specifically, the cradle 106 defines a chamber 142 (FIG. 10) that is configured to receive the power source 14 and an electrical contact 144 that is configured for engagement with one or more corresponding electrical contacts on the power source 14. To direct the UAV 10 during docking and facilitate a proper electrical interface between the power source 14 and the cradle 106, the cradle 106 defines side-walls 146 that taper inwardly towards the chamber 142 so as to define a guide surface 148, as seen in FIG. 10.

The cradle 106 is movable between a retracted position (FIGS. 2, 10), in which the cradle 106 is positioned within the enclosure 102 (e.g., within the internal cavity 112), an extended position (FIG. 13), in which the cradle 106 is positioned externally of the enclosure 102 to facilitate docking with the UAV 10. To support movement between the extended and retracted positions, the cradle 106 is connected to a telescoping slide mechanism 150 that extends through a window 152 (FIG. 1) defined by a bezel portion 154 of the forward frame 116. The slide mechanism 150 supports a cap (hatch, cover) 156 (FIG. 11) that is configured in correspondence with the window 152 such that, upon movement of the cradle 106 into the retracted position, the cap 156 engages the bezel portion 154 and is received within the window 152.

In certain embodiments of the disclosure, such as that illustrated throughout the figures, the cap 156 may include an upstanding tab 158 (FIG. 11) that is configured for receipt within a corresponding recess 160 (FIGS. 8, 12) defined in an inner surface 162 of the door 104. Receipt of the tab 158 within the recess 160 not only allows for proper closure of the door 104, but allows for overlayment of the cap 156 by the door 104 upon closure to facilitate sealing of the enclosure 102 and inhibit (if not entirely prevent) the entry of dust, debris, etc., through the door 104 and/or the bezel portion 154. Additionally, the engagement between the bezel

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portion 154, the slide mechanism 150, and the door 104 (e.g., reception of the tab 158 by the recess 160) improves the security of the base station 100 by inhibiting access to the cradle 106 and, thus, the UAV 10.

In certain embodiments of the disclosure, it is envisioned that the door 104 and the cradle 106 may be automatically actuated upon the receipt of an incoming/docking signal from the UAV 10. For example, it is envisioned that the incoming/docking signal may automatically engage the actuator(s) 118 to thereby open the door 104. Thereafter, when it is determined that the door 104 is fully opened, which may be achieved through the employ of Hall sensors (or any other such suitable detection mechanism), the cradle 106 may be extended via (telescopic) movement of the slide mechanism 150.

Anti-Theft and Security Measures

To further improve the security of the base station 100, in certain embodiments of the disclosure, it is envisioned that the door 104 may include a locking mechanism. For example, it is envisioned that the door 104 may include a magnetic lock to maintain closure of the door 104 in the absence of power to prevent inadvertent and/or unauthorized opening of the door 104 and, thus, access to the UAV 10.

Additionally, or alternatively, it is envisioned that the actuator(s) 118 (FIGS. 8-10) may be configured to resist the application of (manual) force to the door 104, and thereby maintain closure of the door 104, by inhibit (if not entirely preventing) the transmission of force from the door 104 to the motor assembly(ies) 120. For example, it is envisioned that the threading defined by the drive screw 124 may include a fine pitch, which would inhibit (if not entirely prevent) the transmission of force applied to the door 104 to the carrier(s) 126, thereby maintaining closure of the door 104. Additionally, or alternatively, it is envisioned that the actuator(s) 118 may include a solenoid (or other such mechanism) that is configured to engage a locking pin (or other such member).

Pedestal

With reference to FIG. 1, in certain embodiments, the base station 100 may be configured for use with a pedestal 164 (or other such free-standing platform or support) to elevate the base station 100 and the UAV 10. Elevation of the base station 100 and the UAV 10 creates free air space that not only reduces turbulence (e.g., propeller wash) during takeoff and landing of the UAV 10, but mitigates the entry of debris (e.g., dust, particulate, etc.) into the base station 100.

It is envisioned that the pedestal 164 and the base station 100 may include corresponding engagement structures (e.g., pins and holes, detents and recesses, ribs and slots, a footing and a channel, etc.) that are configured for releasable engagement (connection) to promote proper alignment of the pedestal 164 and the base station 100 and inhibit (if not entirely prevent) unintended separation of the base station 100 from the pedestal 164, such as, for example, in the event that the base station 100 and/or the pedestal 164 is subjected to an applied force (e.g., a wind gust, impact with an external object, etc.). In one particular embodiment, it is envisioned that the corresponding engagement structures may include one or more openings and corresponding mechanical fasteners (e.g., bolts, screws, pins, etc.) that are configured for insertion into the opening(s) to allow for fixed, releasable connection of the pedestal 164 and the base station 100.

To inhibit (if not entirely prevent) unauthorized separation of the base station 100 from the pedestal 164 (e.g., to guard against theft of the base station 100 and/or the pedestal 164), it is envisioned that the pedestal 164 and the base station 100

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may include corresponding eyelets (or other such openings) that are configured to receive a lockable tether, chain, cable, bar, etc.

Takeoff, Landing, and Docking

With reference now to FIGS. 14-18 as well, the base station 100 includes a plurality of systems, components, and features that support takeoff of the UAV 10, landing of the UAV 10, and docking of the UAV 10. For example, as described below, the base station 100 may include: one or more connection antennas 166; one or more fiducials 168 that are supported by the enclosure 102; an illumination system 170 that is supported by the enclosure 102; one or more status indicators 172 that are supported by the enclosure 102; and a visualization system 174 that is supported by the enclosure 102.

Connection Antennas

The base station 100 includes primary connection antenna(s) 166 that facilitate wireless communication between the base station 100 and the UAV 10, either directly or indirectly. For example, it is envisioned that the primary connection antenna(s) 166 may be utilized to facilitate communication between the base station 100 and an intervening communication point, such as a hangar, a warehouse, etc. In such embodiments, the primary connection antenna(s) 166 on the base station 100 support direct communication with the hangar (or the like), which would communicate directly with the UAV 10.

In the particular embodiment seen in FIG. 14, for example, the base station 100 includes a pair of primary connection antennas 166 that are secured to (or otherwise engaged with) a roof section 176 of the outer housing 110. It should be appreciated, however, that the particular number of primary connection antennas 166 and/or the location of the primary connection antennas 166 may be varied in alternate embodiments without departing from the scope of the present disclosure. For example, embodiments including a single primary connection antenna 166 are also envisioned herein, as are embodiments in which the base station 100 may include one or more primary connection antennas 166 that are located on sidewalls of the outer housing 110.

In certain embodiments of the disclosure, it is envisioned that the base station 100 may include one more secondary communication antennas that facilitate communication over cellular and/or WiFi networks and/or support GPS functionality. In such embodiments, it is envisioned that the primary connection antenna(s) 166 and the secondary communication antenna(s) may operate in tandem. For example, embodiments are envisioned in which the primary communication antenna(s) 166 may facilitate docking of the UAV 10 with the base station 100 (and communication therebetween) while the secondary communication antenna(s) may facilitate communication between the base station 100 and the hangar (or vice versa).

Fiducials

The fiducials 168 facilitate not only visual identification of the base station 100 by the UAV 10, but guidance of the UAV 10 during landing and docking with the base station 100. To promote or otherwise enhance visualization and/or recognition of the fiducials 168, it is envisioned that the outer housing 110 may include contrasting coloration.

In the particular embodiment of the disclosure seen in FIGS. 13-15, for example, the base station 100 includes (first and second) fiducials 168i, 168ii that are located on (supported by) the roof section 176 of the outer housing 110 (e.g., between the primary connection antennas 166) and a (third) fiducial 168iii that is associated with (e.g., located on or adjacent to) the cradle 106. It should be appreciated,

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however, that the particular number of fiducials **168** and/or the location(s) thereof, may be varied in alternate embodiments without departing from the scope of the present disclosure.

The fiducial **168i** is initially recognized by the UAV **10** to guide the UAV **10** during approach. The fiducial **168i** defines a (first) surface area, which is sufficiently large to allow for visual recognition by the UAV **10** from a desired distance. For example, in certain embodiments, it is envisioned that the (first) surface area defined by the fiducial **168i** may lie substantially within the range of (approximately) 40 percent to (approximately) 80 percent of the surface area defined by the roof section **176**. Surface areas for the fiducial **168i** that lie outside the disclosed range, however, would not be beyond the scope of the present disclosure (e.g., to account for advancements in visualization technology utilized in the UAV **10**).

The fiducial **168ii** is configured as an identification member **178** that is recognized by the UAV **10** during approach to the base station **100** (e.g., after recognition of the fiducial **168i**), which allows the UAV **10** to distinguish amongst a plurality of base stations **100** to facilitate proper pairing (e.g., docking of the UAV **10** to a specific base station **100**). In certain embodiments of the disclosure, it is envisioned that the fiducial **168i** may be fixedly connected to the base station **100** (e.g., the roof section **176**) and that the fiducial **168ii** may be configured for removable connection to the base station **100** (e.g., the roof section **176**). Removable connection of the fiducial **168ii** allows for the uniform manufacture of a fleet of base stations **100** and the subsequent attachment of the fiducials **168ii** thereto. Embodiments in which the fiducial **168ii** may be integrally (e.g., monolithically) formed with the base station **100**, however, would not be beyond the present disclosure.

In the particular embodiment illustrated, the fiducial **168ii** defines a (second) surface area, which is less than the (first) surface area defined by the fiducial **168i**. For example, it is envisioned that the (second) surface area defined by the fiducial **168ii** may lie substantially within the range of (approximately) 10 percent to (approximately) 50 percent of the (first) surface area defined by the fiducial **168i**. Surface areas for the fiducial **168ii** that lie outside the disclosed range, however, would not be beyond the scope of the present disclosure (e.g., to account for advancements in visualization technology utilized in the UAV **10**).

The fiducial **168iii** is configured as an April tag **180** and is recognized by the UAV **10** after recognition of the fiducial **168ii**. In the particular embodiment of the disclosure illustrated, the fiducial **168iii** is located on the slide mechanism **150**, which inhibits (if not entirely prevents) any interference with air flow across the cradle **106** during cooling and/or heating of the power source **14** of the UAV **10**, which is discussed in further detail below. Embodiments in which the fiducial **168iii** may be located on the cradle **106** itself, however, are also envisioned herein and would not be beyond the scope of the present disclosure.

Illumination System

The illumination system **170** is configured to improve visibility of the fiducials **168** during nighttime operation. In the particular embodiment of the disclosure illustrated, the illumination system **170** includes one or more light sources **182** (FIG. 15) (e.g., LEDs **184**) that are secured, mounted to, or otherwise supported by the roof section **176** of the outer housing **110** to support lighting (illumination) of the fiducials **168i**, **168ii**. It is envisioned that the illumination system **170** may be connected to any suitable power source, whether internal to the base station **100** (e.g., to the power supply

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controlled by the main board/processor) or external (e.g., to a separate power supply, battery, or the like).

In certain embodiments of the disclosure, it is envisioned that the illumination system **170** may be controlled by the main board/processor and configured to flash or strobe the light source(s) **182** according to a particular pattern, which can be recognized by the UAV **10** during approach to thereby identify the base station **100**. In such embodiments, it is envisioned that the illumination system **170** may either supplement or replace the fiducial **168ii** as a means of identifying the base station **100**.

Status Indicators

The status indicator(s) **172** (FIG. 16) identify the status of the base station **100** (e.g., that the base station **100** is ready, requires service, is undergoing the docking procedure with the UAV **10**, etc.). In certain embodiments, it is envisioned that the status indicator(s) **172** may include a series of LEDs **186**, which not only supports status indication, but illumination of the fiducial **168iii** (FIG. 13) and/or the cradle **106** (e.g., to facilitate docking of the UAV **10**). For example, in one particular embodiment, it is envisioned that one or more of the LEDs **186** may be configured to emit a colored light (e.g., red, blue, yellow, etc.) and that one or more of the LEDs **186** may be configured to emit a white light to illuminate the fiducial **168iii** and/or the cradle **106**.

In the particular embodiment of the disclosure illustrated in FIG. 16, the status indicator(s) **172** are secured, mounted to, or otherwise supported by the forward frame **116** of the outer housing **110**. It should be appreciated, however, that the particular location of the status indicator(s) **172** may be varied in alternate embodiments without departing from the scope of the present disclosure. It is envisioned that the status indicator(s) **172** may be connected to any suitable power source, whether internal to the base station **100** (e.g., to the power supply controlled by the main board/processor) or external (e.g., to a separate power supply, battery, or the like).

UAV Storage

In certain embodiments of the disclosure, it is envisioned that the propulsion mechanism(s) **12** (FIG. 2) (e.g., the propellers **18**) of the UAV **10** may be reconfigurable between unfolded (flight) and folded (storage) configurations, which allows for a reduction in the overall size of the base station **100**. To support movement from the unfolded configuration to the folded configuration, it is envisioned that the base station **100** may be configured for engagement with the of the propulsion mechanism(s) **12** (e.g., during retraction of the cradle **106**). For example, it is envisioned that the base station **100** may include one or more engagement members that are configured to physically interface with the propulsion mechanism(s) **12** to move the propulsion mechanism(s) **12** into the folded configuration. It is envisioned that the engagement member(s) may be integrated into (or defined by) one or more surface(s) of the enclosure **102** (e.g., the forward frame **116** of the outer housing **110**). For example, the engagement member(s) may be configured as rollers, brushes, or the like.

In certain embodiments of the disclosure, it is envisioned that the engagement member(s) may be configured for passive interaction with the propulsion mechanism(s) **12**, whereby the propulsion mechanism(s) **12** are brought into contact with the engagement member(s) by virtue of the retraction of the cradle **106**. Alternatively, it is envisioned that the engagement member(s) may be configured for active engagement (interaction) with the propulsion mechanism(s) **12**. For example, it is envisioned that the engagement member(s) may be repositionable (reconfigurable) between

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a first position (configuration), in which the engagement member(s) are positioned (configured) to avoid contact with the propulsion mechanism(s) 12 (e.g., such that the engagement member(s) are located outside the path followed by the propulsion mechanism(s) 12 during retraction of the cradle 106), and a second position (configuration), in which the engagement member(s) are positioned (configured) for contact with the propulsion mechanism(s) 12 (e.g., such that the engagement member(s) are located within the path followed by the propulsion mechanism(s) 12 during retraction of the cradle 106).

To allow for additional reductions in the overall size of the base station 100, it is envisioned that the base station 100 may include one or more contact members 188 that are configured for engagement (contact) with one or more antennas 34 on the UAV 10 to facilitate reconfiguration (repositioning, folding) thereof between an active (use, unfolded, deployed) configuration (FIG. 2), in which the antenna(s) 34 extend outwardly (away) from a body 36 of the UAV 10, which supports (is connected to) the power source 14, and a passive (storage, folded, undeployed) configuration, in which the antenna(s) 34 are positioned adjacent to (e.g., in contact (engagement) with the body 36 of the UAV 10). Reconfiguration of the antenna(s) 34 from the active configuration into the passive configuration creates clearance with the internal components of the base station 100 to inhibit (if not entirely prevent) unintended contact between the UAV 10 and the base station 100 during movement of the cradle 106 (and the UAV 10) between the extended position (FIG. 13) and the retracted position (FIGS. 2, 10). In the particular embodiment of the UAV 10 illustrated throughout the figures, the antenna(s) 34 are biased towards the active configuration via one or more biasing members (e.g., springs or the like) such that the antenna(s) 34 are automatically reconfigured from the passive configuration into the active configuration as the UAV 10 exits the base station 100 (e.g., via movement of the cradle 106 from the retracted position (FIGS. 2, 10) into the extended position (FIG. 13)).

In the embodiment illustrated in FIG. 18A, the contact member 188 defines a groove (channel) 189 that is configured to receive the antenna(s) 34 during retraction of the cradle 106 as the UAV 10 is withdrawn into the enclosure 102. FIG. 18B illustrates another embodiment of the contact member 188 that includes a leading (forward) edge 190 and a trailing (rear) edge 192 and defines a (vertical) height H that varies between the edges 190, 192 so as to define an angled (chamfered, beveled) surface 194 that is configured for engagement (contact) with the antenna(s) 34 on the UAV 10. In the particular embodiment of the disclosure seen in FIG. 18B, each contact member 188 tapers such that the height H decreases from the leading edge 190 towards the trailing edge 192. Embodiments in which each contact member 188 may taper such that the height H increases from the leading edge 190 towards the trailing edge 192 are also envisioned herein and would not be beyond the scope of the present disclosure (e.g., depending upon the particular configuration and/or location of the antenna(s) on the UAV 10, spatial constraints of the enclosure 102, etc.).

In the embodiments illustrated in FIGS. 18A and 18B, the base station 100 includes a single contact member 188 that is secured to (or otherwise engaged with) an inner (e.g., upper) surface 196 of the inner housing 108. It should be appreciated, however, that the particular number of contact members 188 and/or the location of the contact member(s) 188 may be varied in alternate embodiments without departing from the scope of the present disclosure (e.g., depending

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upon the number of antennas 34 on the UAV 10 and/or the location thereof). For example, embodiments including multiple contact members 188 are also envisioned herein, as are embodiments in which the base station 100 may include one or more contact members 188 that are located on sidewalls 198 of the inner housing 108.

It is envisioned that the contact member(s) 188 may be secured to the inner housing 108 in any suitable manner. For example, it is envisioned that the contact member(s) 188 may be integrally (e.g., monolithically) formed with the inner housing 108 or that the contact member(s) 188 and the inner housing 108 may be formed as separate, discrete structures, which may be secured together via one or more mechanical fasteners, an adhesive, etc.

15 Temperature Control System

With reference now to FIGS. 19-24, the base station 100 includes a temperature control (e.g., heating and cooling) system 200 that is connected to the cradle 106 and is configured to vary (regulate) the temperature of the power source 14 (FIGS. 2, 4, 5) when the UAV 10 is docked in the base station 100 (e.g., cool or heat the UAV 10 subject to environmental conditions). In the particular embodiment of the disclosure illustrated throughout the figures, the temperature control system 200 is configured to cool the power source 14 of the UAV 10 (e.g., when the UAV 10 and the base station 100 are operated in warmer environments). It should be appreciated, however, that embodiments in which the temperature control system 200 may be configured to heat the power source 14 of the UAV 10 (e.g., when the UAV 10 and the base station 100 are operated in cooler environments) are also envisioned herein and would not be beyond the scope of the present disclosure, as described in further detail below.

The temperature control system 200 includes: an upper air circuit 202; a lower air circuit 204; and a thermoelectric conditioner (TEC) 206 that is thermally connected to, and located between, the respective upper and lower air circuits 202, 204.

The upper air circuit 202 is an open system that receives and circulates ambient air, which may be sourced from within the base station 100 or externally of the base station 100 (e.g., via an air intake) to vary (regulate) the temperature of the TEC 206. The upper air circuit 202 includes: an upper (first) plenum 208; an upper (first) air circulator 210; and an upper (first) heat sink 212.

The upper plenum 208 includes an (upper) ducting system 214 that receives and circulates the ambient air. The ducting system 214 may include any suitable material (or combination of materials) and may include either a unitary configuration, in which the upper plenum 208 is formed from a single piece of material, or a segmented configuration, in which the upper plenum 208 is formed from a plurality of segments that are connected together (e.g., via one or more mechanical fasteners, an adhesive, in an interference fit, etc.).

The upper air circulator 210 supports and directs air flow through the upper plenum 208 and across the upper heat sink 212 to vary air temperature within the upper air 202 and thereby regulate the temperature of the TEC 206, as described in further detail below. The upper air circulator 210 may include any structure or mechanism suitable for that intended purpose and may be positioned in any location suitable. For example, in the illustrated embodiment, the upper air circulator 210 is configured as a fan 216 that is located within the upper plenum 208 (e.g., within the ducting system 214). It is also envisioned, however, that the upper air circulator 210 may be located externally of the

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upper plenum 208. For example, the upper air circulator 210 may be connected to (or otherwise supported on) an exterior surface 218 of the ducting system 214.

The upper heat sink 212 (FIGS. 23, 24) is connected to (e.g., located within) the upper plenum 208 and is configured to alter the temperature of the air circulated through the upper air circuit 202. For example, when the temperature control system 200 is utilized to cool the power source 14 (FIGS. 2, 4, 5) of the UAV 10, the upper heat sink 212 absorbs and distributes thermal energy (heat) generated by the TEC 206 to lower the temperature of the air flowing through the upper plenum 208, as described in further detail below. To increase the absorption and distribution of thermal energy by the upper heat sink 212, air flow through the upper plenum 208 and, thus, air flow across the upper heat sink 212, may be increased by increasing the speed of the upper air circulator 210 (e.g., by increasing power to the fan 216).

The lower air circuit 204 includes a lower (second) plenum 220; a lower (second) air circulator 222; and a lower (second) heat sink 224 and is configured as a closed system. As such, in contrast to the upper air circuit 202, rather than drawing in additional ambient air, the lower air circuit 204 continuously circulates the air that is naturally present within the lower plenum 220.

The lower plenum 220 includes a (lower) ducting system 226 that directs air flow across the cradle 106 and the power source 14 of the UAV 10 when the UAV 10 is docked within the base station 100 and may include any suitable material (or combination of materials). The lower plenum 220 (e.g., the ducting system 226) includes a segmented (non-unitary) configuration that defines a rear (first, fixed) section 228 and a forward (second, movable) section 230 that is movable in relation to the rear section 228. The rear section 228 of the lower plenum 220 is fixedly connected (secured) to the TEC 206 and, as such, is fixed in relation to the upper air circuit 202. The forward section 230 of the lower plenum 220 is connected (secured) to the cradle 106 and is movable therewith during repositioning of the cradle 106 between the retracted position (FIGS. 2, 10) and the extended position (FIG. 13). The forward section 230 of the lower plenum 220 receives air from the rear section 228 and defines an air inlet 232 and an air outlet 234. The air inlet 232 and the air outlet 234 each include one or more slits (or other such openings) that extend through the cradle 106 (e.g., the sidewalls 146) and into the chamber 142, which allows for air cooled by the lower air circuit 204 to be directed into the chamber 142 and across the power source 14 of the UAV 10 when the UAV 10 is docked in the base station 100. More specifically, the air inlet 232 is configured to direct air into the cradle 106 and across the power source 14 of the UAV 10 and the air outlet 234 is configured to receive the air directed across the power source 14 and redirect the air through the lower plenum 220 and across the lower heat sink 224, as described in further detail below.

Although shown as extending entirely about the cradle 106 in the particular embodiment illustrated throughout the figures, embodiments are also envisioned in which the forward section 230 of the lower plenum 220 may only partially circumscribe the cradle 106. For example, embodiments are envisioned in which the forward section 230 of the lower plenum 220 may include opposite terminal ends that respectively define the air inlet 232 and the air outlet 234.

Additionally, while the forward section 230 of the lower plenum 220 and the cradle 106 are illustrated as being integrally (e.g., monolithically) formed in the illustrated embodiment, it is also envisioned that the lower plenum 220 and the cradle 106 may be configured for releasable (detach-

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able) engagement to allow for repeated connection and disconnection of the lower plenum 220 and the cradle 106 (e.g., via corresponding engagement structures such as detents, clips, fasteners, or the like).

The lower air circulator 222 supports and directs air flow through the lower plenum 220 and across the lower heat sink 224 to thermally condition the air within the lower air circuit 204 (e.g., vary the temperature thereof) and thereby heat or cool the power source 14 of the UAV 10, as described in further detail below. The lower air circulator 222 may include any structure or mechanism suitable for that intended purpose and may be positioned in any location suitable. For example, in the illustrated embodiment, the lower air circulator 222 is configured as a fan 236 that is located within the lower plenum 220 (e.g., within the ducting system 226). It is also envisioned, however, that the lower air circulator 222 may be located externally of the lower plenum 220. For example, the lower air circulator 222 may be connected to (or otherwise supported on) an exterior surface 238 of the ducting system 226.

The lower heat sink 224 is connected to (e.g., located within) the lower plenum 220 and is configured to treat the air circulated therethrough (e.g., alter the temperature of the air within the lower plenum 220 via cooling or heating) prior to direction across the power source 14 of the UAV 10. For example, when the temperature control system 200 is utilized to cool the power source 14 of the UAV 10, the lower heat sink 224 absorbs and distributes thermal energy (heat) from the air flowing through the lower plenum 220 to lower the temperature thereof and, thus, the power source 14 of the UAV 10. To increase the absorption and distribution of thermal energy by the lower heat sink 224 and, thus, enhance cooling of the power source 14 of the UAV 10, air flow through the lower plenum 220 and, thus, air flow across the lower heat sink 224, may be increased by increasing the speed of the lower air circulator 222 (e.g., by increasing power to the fan 236).

The TEC 206 is configured as a Peltier system and includes a dedicated/integrated power source/control as well as a first (upper, "hot") end 240 (FIG. 23) and a second (lower, "cold") end 242, each of which includes a thermal member (e.g., a ceramic plate). The first end 240 is thermally and/or physically connected to the upper heat sink 212 and the second end 242 is thermally and/or physically connected to the lower heat sink 224. Upon activation, as current flows through the TEC 206, the temperature of the first end 240 increases (to an upper temperature threshold) while the temperature of the second end 242 decreases (to a lower temperature threshold) until a predetermined, fixed temperature differential is realized. For example, it is envisioned that the differential between the respective upper and lower ends 240, 242 of the TEC 206 may lie substantially within the range of (approximately) 30° C. to (approximately) 70° C. (e.g., (approximately) 50° C.). Embodiments in which the temperature differential may lie outside the disclosed range, however, are also envisioned herein and would not be beyond the scope of the present disclosure. Consequently, reducing the upper temperature threshold allows for a corresponding reduction in the lower temperature threshold.

During operation of the temperature control system 200, in the particular embodiment of the disclosure illustrated, thermal energy (heat) generated by the TEC 206 is absorbed and dissipated by the upper heat sink 212 and the ambient air flowing through the upper plenum 208. The upper air circuit 202 thus cools the first ("hot") end 240 of the TEC 206, which results in corresponding cooling of the second ("cold") end 242 of the TEC 206 and, thus, increased

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cooling of the air flowing through the lower plenum **220** and the power source **14** of the UAV **10** when the UAV **10** is docked in the base station **100**.

Although illustrated as including a single TEC **206** in the particular embodiment shown throughout the figures, embodiments are also envisioned in which the temperature control system **200** may include multiple TECs **206**. In such embodiments, it is envisioned that the TECs **206** may be identical or non-identical in configuration (e.g., it is envisioned that the temperature control system **200** may include TECs **206** that vary in size) and/or that the TECs **206** may be arranged in series or in parallel (e.g., in a stacked configuration).

As indicated above, embodiments of the disclosure are envisioned herein in which the temperature control system **200** may be configured to heat, rather than cool, the power source **14** of the UAV **10** when the UAV **10** is docked in the base station **100**. In such embodiments, current flow through the TEC **206** can be reversed (e.g., via electronic control) such that the first end **240** of the TEC **206** functions as the “cold” end and the second end **242** of the TEC **206** functions as the “hot” end.

During use of the temperature control system **200**, the upper air circuit **202** draws air in from the ambient, either from within the base station **100** or externally of the base station **100**, which is directed across the upper heat sink **212** via the upper air circulator **210**. As the air flows across the upper heat sink **212**, heat is withdrawn, thereby cooling the upper heat sink **212** and, thus, the first (“hot”) end **240** of the TEC **206**, and heating the air. The heated air is then discharged from the upper air circuit **202**, being expelled either into the base station **100** or externally thereof (e.g., through a vent), and is replaced by cooler ambient air that is drawing into the upper air circuit **202** by the upper air circulator **210**.

Upon docking of the UAV **10**, as the cradle **106** moves from the extended position (FIG. **13**) into the retracted position (FIGS. **2**, **10**), the forward section **230** of the lower plenum **220** mates with (engages) the rear section **228**, which closes the lower air circuit **204** and allows for the continuous circulation of air therethrough. To facilitate mating engagement between the forward section **230** and the rear section **228**, it is envisioned that the forward section **230** and the rear section **228** may include one or more seals (e.g., O-rings) or corresponding engagement structures (e.g., collars, flanges, or the like) at the interface therebetween.

Air flowing through the lower air circuit **204** is directed across the lower heat sink **224** via the lower air circulator **222**, which withdraws heat from the air to cool the air prior to entering the chamber **142** of the cradle **106** via the air inlet **232**. As the cooled air flows through across the cradle **106** and through the chamber **142**, heat is withdrawn from the power source **14** (FIGS. **2**, **4**, **5**) of the docked UAV **10**, which is facilitated by the heat exchanger **24**. More specifically, in the illustrated embodiment, the cooled air flows through the channels **32** (FIG. **4**) defined by the diffusers **26** (e.g., the fins **30**), which increases the distribution of thermal energy from the power source **14** of the UAV **10** to enhance cooling, and increases the temperature of the air. After flowing across the power source **14**, the (heated) air exits the chamber **142** via the air outlet **234** and is redirected across the lower heat sink **224** (e.g., via the lower air circulator **222**) to again cool the air prior to recirculation through the cradle **106** and across the power source **14**.

Weather and Climate Management

To allow for operation in various weather conditions, the base station **100** includes a plurality of components and

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systems that are configured to regulate temperature, moisture, humidity, and the like in order to maximize operability in a variety of environments.

Snow and Ice

As seen in FIG. **25**, in certain embodiments of the disclosure, the base station **100** includes one or more heating elements **244** that are thermally and/or physically connected to (supported by) the enclosure **102** (e.g., via an adhesive). Although shown as being associated with (supported by) the roof section **176** of the outer housing **110** in the particular embodiment illustrated, it is envisioned that the heating element(s) **244** may be thermally and/or physically connected to any section(s) of the outer housing **110** that may benefit from heating. Upon activation, the heating element(s) **244** increase the temperature of (heat) the outer housing **110** (e.g., to reduce the presence of snow and/or ice on the roof section **176**).

It is envisioned that the heating element(s) **244** may be connected to any suitable power source, whether internal to the base station **100** (e.g., to the power supply controlled by the main board/processor) or external (e.g., to a separate power supply, battery, or the like), and that the heating element(s) **244** may be either manually or automatically activated. For example, it is envisioned that the heating element(s) **244** may be activated via a signal that is relayed by one or more temperature sensors **246** that are in communication with the heating element(s) **244** and which are configured to detect when the temperature crosses (e.g., falls below or exceeds) a predetermined threshold (e.g., 32° F.). Additionally, or alternatively, it is envisioned that the heating element(s) **244** may be activated upon receiving an activation signal from a weather station (e.g., via a cloud-based connection) and/or from the visualization system **174**, which may be configured to visually detect the presence of snow and/or ice.

Internal Temperature and Humidity Regulation

To regulate (control) the temperature and/or humidity within the base station **100**, it is envisioned that the base station **100** may include one or more internal fans **248** (FIG. **26**), which may be supported in any suitable location on the inner housing **108** and/or the outer housing **110**. The internal fan(s) **248** regulate (e.g., vary, control) temperature and/or humidity within the base station **100** and may be configured to either cool the base station **100** or heat the base station **100** (e.g., via thermal connection to one or more heating elements or components).

The internal fan(s) **248** are controllable (e.g., via the main board/processor) to draw air in and exhaust air through one or more ports/vents in the outer housing **110** and/or the inner housing **108**, the location(s) of which may be varied to direct air flow in a particular direction (e.g., across the UAV **10**). For example, it is envisioned that the port(s)/vent(s) may be located and/or configured to create air flow through the base station **100** in any effective (or otherwise desired) pattern.

In certain embodiments of the disclosure, it is envisioned that the internal fan(s) **248** may be automatically activated via a signal that is relayed by one or more sensors **250** that are configured to detect temperature, humidity, etc. Additionally, or alternatively, it is envisioned that the internal fan(s) **248** may be connected to a timer such that the internal fan(s) **248** are automatically activated at a particular time of day.

In the context of humidity regulation, upon the detection of moisture, the sensor(s) **250** may generate an activation signal that can be utilized to initiate one or more mitigation processes. For example, it is envisioned that the sensor(s) may be in communication with the internal fan(s) **248** such

that the internal fan(s) **248** are engaged upon receipt of the activation signal from the sensor(s) **250** to remove (or otherwise mitigate) excess humidity within the base station **100**, thereby inhibiting (if not entirely preventing) condensation that might otherwise compromise the functionality of one or more components of the UAV **10** or the base station **100**. For example, the presence of condensation may result in malfunction and/or damage to the electronics module (e.g., the main board/processor) and/or “fogging” of the visualization system **174**. To further inhibit (if not entirely prevent) the presence of humidity, condensation, moisture, etc., in certain embodiments of the disclosure, it is envisioned that the electronics module may be sealed within the base station **100**. For example, the electronics module, or the various components thereof (e.g., motor drivers, interface boards, lighting boards, etc.), may be sealed, either collectively (via hermetic sealing) or individually (e.g., via dip coating).

Drainage

In certain embodiments of the disclosure, the enclosure **102** (e.g., the outer housing **110**) may include one or more channels **252** (FIG. 27) that are configured to collect and direct water away from any entry points into the enclosure **102** in a manner that inhibits (if not entirely prevents) entry into the base station **100** and/or away from any components, whether electronic, mechanical, or otherwise, that may be compromised by the presence of moisture. For example, it is envisioned that the channel(s) **252** may be configured to collect and direct water away from the interface with any power cables, any locking members/mechanisms, the door **104** (FIG. 1), etc. While the enclosure **102** is shown as including a single channel **252** that extends about a rear periphery **254** of the outer housing **110** in the particular embodiment illustrated, it should be appreciated that the number of channels **252** and/or the location of the channels **252** may be varied without departing from the present disclosure. For example, embodiments including one or more additional channels **252** are also envisioned herein, as are embodiments in which the enclosure **102** may include one or more channels **252** that extend about a front periphery **256** (FIG. 1) of the outer housing **110**, the roof section **176**, etc.

In the particular embodiment of the base station **100** illustrated in FIG. 27, the channel **252** is defined by an (molded) insert **258** that is incorporated into the outer housing **110**. It is envisioned, however, that the channel(s) **252** may be configured in any manner suitable for the intended purpose of collecting and/or directing water in the manner described herein. For example, embodiments are also envisioned in which the channel(s) **252** may be integrally (e.g., monolithically) formed with the outer housing **110**.

In certain embodiments of the disclosure, to further inhibit (if not entirely prevent) water penetration, it is envisioned that the enclosure **102** may include one or more seals, gaskets, etc., that are associated with the channel(s) **252**. For example, it is envisioned that such seals, gaskets, etc., may be positioned about the insert **258** and supported by the inner housing **108** and/or the outer housing **110**.

It should be appreciated that any of the aforementioned components and/or systems may be omitted in order to reduce the overall cost and complexity of the base station **100**. For example, in hot (e.g., desert) climates, it is envisioned that the heating element(s) **244** may be eliminated.

Visualization System

With reference now to FIG. 1, the visualization system **174** includes a digital image capturing device **260** (e.g., a

digital camera) or the like. It is envisioned that the visualization system **174** may be connected to any suitable power source, whether internal to the base station **100** (e.g., to the power supply controlled by the main board/processor) or external (e.g., to a separate power supply, battery, or the like).

In the particular embodiment of the disclosure illustrated, the visualization system **174** includes a single digital image capturing device **260** that is secured, mounted to, or otherwise supported by the forward frame **116** of the outer housing **110**, which supports observation and visual analysis of the environment in which the base station **100** is located as well as observation and visual analysis of the UAV **10** (FIG. 2) prior to takeoff, during takeoff, and during landing. It should be appreciated, however, that the number of digital image capturing devices **260** and/or the location of the digital image capturing device(s) **260** may be varied in alternate embodiments without departing from the present disclosure. For example, embodiments including one or more additional digital image capturing devices **260** are also envisioned herein.

During operation of the base station **100**, the visualization system **174** supports visual inspection of the environment, which not only improves safety of the base station **100** and the UAV **10** by confirming the absence of people, animals, etc., prior to takeoff, during takeoff, and during landing of the UAV **10**, but functionality of the base station **100** as well. For example, it is envisioned that the visualization system **174** may be configured to identify precipitation (e.g., snow, ice, rain, etc.) and actuate (trigger) operation of the heating element(s) **244** (FIG. 25), the internal fan(s) **248** (FIG. 26), or other such systems. It is also envisioned that the visualization system **174** may be utilized to inspect the UAV **10** (e.g., prior to takeoff and/or during docking) and identify any damage that may result in subsequent suboptimal performance.

In certain embodiments, the one or more diffusers extend continuously between opposite ends of the power source.

In certain embodiments, the one or more antennas include first ends connected to an exterior surface of the body and second free ends.

While the present disclosure has been described in connection with certain embodiments, it is to be understood that the present disclosure is not to be limited to the disclosed embodiments, but, on the contrary, is intended to cover various modifications and equivalent arrangements included within the scope of the appended claims, which scope is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures as is permitted under the law.

Persons skilled in the art will understand that the various embodiments of the present disclosure and shown in the accompanying figures constitute non-limiting examples, and that additional components and features may be added to any of the embodiments discussed hereinabove without departing from the scope of the present disclosure. Additionally, persons skilled in the art will understand that the elements and features shown or described in connection with one embodiment may be combined with those of another embodiment without departing from the scope of the present disclosure to achieve any desired result and will appreciate further features and advantages of the presently disclosed subject matter based on the description provided. Variations, combinations, and/or modifications to any of the embodiments and/or features of the embodiments described herein that are within the abilities of a person having ordinary skill in the art are also within the scope of the present disclosure,

as are alternative embodiments that may result from combining, integrating, and/or omitting features from any of the disclosed embodiments.

Use of the term “optionally” with respect to any element of a claim means that the element may be included or omitted, with both alternatives being within the scope of the claim. Additionally, use of broader terms such as “comprises,” “includes,” and “having” should be understood to provide support for narrower terms such as “consisting of,” “consisting essentially of,” and “comprised substantially of.” Accordingly, the scope of protection is not limited by the description set out above, but is defined by the claims that follow, and includes all equivalents of the subject matter of the claims.

In the preceding description, reference may be made to the spatial relationship between the various structures illustrated in the accompanying drawings, and to the spatial orientation of the structures. However, as will be recognized by those skilled in the art after a complete reading of this disclosure, the structures described herein may be positioned and oriented in any manner suitable for their intended purpose. Thus, the use of terms such as “above,” “below,” “upper,” “lower,” “inner,” “outer,” “left,” “right,” “upward,” “downward,” “inward,” “outward,” “horizontal,” “vertical,” etc., should be understood to describe a relative relationship between the structures and/or a spatial orientation of the structures. Those skilled in the art will also recognize that the use of such terms may be provided in the context of the illustrations provided by the corresponding figure(s).

Additionally, terms such as “approximately,” “generally,” “substantially,” and the like should be understood to allow for variations in any numerical range or concept with which they are associated. For example, it is intended that the use of terms such as “approximately” and “generally” should be understood to encompass variations on the order of 25%, or to allow for manufacturing tolerances and/or deviations in design.

Although terms such as “first,” “second,” “third,” etc., may be used herein to describe various operations, elements, components, regions, and/or sections, these operations, elements, components, regions, and/or sections should not be limited by the use of these terms in that these terms are used to distinguish one operation, element, component, region, or section from another. Thus, unless expressly stated otherwise, a first operation, element, component, region, or section could be termed a second operation, element, component, region, or section without departing from the scope of the present disclosure.

Each and every claim is incorporated as further disclosure into the specification and represents embodiments of the present disclosure. Also, the phrases “at least one of A, B, and C” and “A and/or B and/or C” should each be interpreted to include only A, only B, only C, or any combination of A, B, and C.

What is claimed is:

1. An unmanned aerial vehicle (UAV) comprising:

a power source including:

one or more power cells;

one or more thermal transfer members thermally connected to the one or more power cells, wherein the one or more power cells and the one or more thermal transfer members correspond in number; and

a heat exchanger thermally connected to the one or more thermal transfer members such that the one or more thermal transfer members extend between the one or more power cells and the heat exchanger, whereby the one or more thermal transfer members

and the heat exchanger facilitate a transfer of thermal energy between the power source and ambient air to decrease or increase temperature of the power source.

2. The UAV claim 1, wherein the one or more thermal transfer members include graphite.

3. The UAV claim 1, wherein the one or more thermal transfer members are unitary in construction.

4. The UAV of claim 1, wherein the one or more power cells include a plurality of individual power cells and the one or more thermal transfer members include a plurality of individual thermal transfer members.

5. The UAV of claim 1, wherein the heat exchanger includes one or more diffusers to increase surface area of the heat exchanger and thermal energy distribution towards or away from the power source.

6. The UAV of claim 5, wherein the one or more diffusers extend axially and/or laterally along an outer surface of the heat exchanger.

7. The UAV of claim 5, wherein the one or more diffusers include a plurality of diffusers.

8. The UAV of claim 7, wherein the plurality of diffusers are configured as fins defining one or more channels therebetween, the one or more channels being configured to direct air flow along the heat exchanger to increase thermal energy distribution towards or away from the power source.

9. An unmanned aerial vehicle (UAV) comprising:

a power source including a heat exchanger configured to transfer thermal energy between the power source and ambient air to decrease or increase temperature of the power source, the heat exchanger including one or more diffusers to increase surface area of the heat exchanger and thermal energy distribution towards or away from the power source, wherein the one or more diffusers extend continuously between opposite ends of the power source.

10. The UAV of claim 9, wherein the one or more diffusers extend axially and/or laterally along an outer surface of the heat exchanger.

11. The UAV of claim 9, wherein the one or more diffusers includes a plurality of diffusers.

12. The UAV of claim 11, wherein the plurality of diffusers are configured as fins defining one or more channels therebetween, the one or more channels being configured to direct air flow along the heat exchanger to increase thermal energy distribution towards or away from the power source.

13. An unmanned aerial vehicle (UAV) comprising:

a body;

one or more antennas extending from the body, the one or more antennas being reconfigurable between an active configuration, in which the one or more antennas extend outwardly from the body, and a passive configuration, in which the one or more antennas are positioned adjacent to the body, wherein the one or more antennas include first ends connected to an exterior surface of the body and second free ends; and

a power source connected to the body, the power source including one or more diffusers extend axially and/or laterally along an outer surface of the power source to transfer thermal energy between the power source and ambient air to decrease or increase temperature of the power source.

14. The UAV of claim 13, wherein the one or more antennas are biased towards the active configuration.

15. The UAV of claim 13, wherein the one or more diffusers includes a plurality of diffusers configured as fins

defining one or more channels therebetween, the one or more channels being configured to direct air flow along the power source to increase thermal energy distribution towards or away from the power source.

16. The UAV of claim 13, wherein the power source 5 further includes:

one or more power cells; and
one or more thermal transfer members thermally connected to the one or more power cells and the one or more diffusers. 10

17. The UAV of claim 16, wherein the one or more thermal transfer members include graphite.

18. The UAV of claim 16, wherein the one or more thermal transfer members are unitary in construction.

19. The UAV of claim 6, wherein the one or more 15 diffusers extend continuously between opposite ends of the power source.

20. The UAV of claim 9, wherein the power source further includes:

one or more power cells; and 20
one or more thermal transfer members thermally connected to the one or more power cells, wherein the heat exchanger is thermally connected to the one or more thermal transfer members such that the one or more thermal transfer members extend between the one or 25 more power cells and the heat exchanger.

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